

JÖNKÖPING UNIVERSITY
School of Engineering

Carry the Forest:

Designing a Bag for Forestry Machine Operators

PAPER WITHIN *Product Design*

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Abstract

Forestry machine operators often need to transport both personal and work-related items between home and the cabin daily. These users face practical challenges due to a lack of organization, poor insulation for food and drinks and limited space within the machine.

This master's thesis explores the design of a multifunctional backpack for forestry machine operators using user-centered research methods such as questionnaire, mock-ups, user testing or material analysis. The design process followed the Double Diamond methodology, where three iterations were conducted. The development and refine of the concept were achieve by iterating through ideation, prototyping, and testing to translate the user insights into practical design features.

The research identified key user needs: compartmentalization, durability, comfort and compatibility with the cabin space, which were all integrated into the backpack design.

The final backpack design includes modular compartments, waterproof and insulating materials, and ergonomic features suited for harsh forestry environments and the operator's needs. It improves organization of the user, protects contents and adapts to the specific challenges users face.

The project was developed in collaboration with Terri AB Skogmaskiner and supports the company's brand identity through a strategic use of colour and materials.

Keywords

Storage, workspace, forestry machine operators, multifunctional bag, transporting belongings, ergonomics, compartmentalization, backpack design, user-centered design.

Sammanfattning

Skogsmaskinsoperatörer behöver ofta dagligen transportera både personliga och arbetsrelaterade föremål mellan hemmet och hytten. Dessa användare står inför praktiska utmaningar på grund av bristande organisering, dålig isolering för mat och dryck samt begränsat utrymme i maskinen.

Denna masteruppsats utforskar designen av en multifunktionell ryggsäck för skogsmaskinsoperatörer genom användarcentrerade forskningsmetoder som enkätundersökningar, mock-ups, användartester och materialanalys. Designprocessen följde Double Diamond-metodologin, där tre iterationer genomfördes. Utvecklingen och förfiningen av konceptet uppnåddes genom idéutveckling, prototypframtagning och testning, för att omvandla användarinsikter till praktiska designfunktioner.

Forskningen identifierade användarnas främsta behov: fackindelning, hållbarhet, komfort och kompatibilitet med utrymmet i hytten, vilka alla integrerades i ryggsäckens design.

Den slutliga ryggsäcksdesignen innehåller modulära fack, vattentäta och isolerande material samt ergonomiska egenskaper anpassade för tuffa skogsmiljöer och operatörernas behov. Den förbättrar användarens organisering, skyddar innehållet och anpassar sig till de specifika utmaningar som användarna möter.

Projektet utvecklades i samarbete med Terri AB Skogsmaskiner och stödjer företagets varumärkesidentitet genom en strategisk användning av färg och material.

Nyckelord

Förvaring, arbetsmiljö, skogsmaskinsoperatörer, multifunktionell väska, transport av tillhörigheter, ergonomi, fackindelning, ryggsäckdesign, användarcentrerad design.

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1. Introduction

This project explains the concept development of a multifunctional bag for forestry machinery professionals in Sweden.

1.1. Background

1.1.1 Forestry work

Sweden is known for its abundant forests. 69% of Sweden's total land area is covered by forest. Thanks to this vast amount of forest area, there is a big industry that utilizes its resources, which makes Sweden the fifth largest exporter of pulp, paper, and sawn timber in the world '. These forestlands represent 24 million hectares of land and are owned by different groups of people. Most of them are owned by individual private owners (48%), followed by private companies (24%), then the Swedish state and municipalities (22%) and lastly other private owners (6%), such as the Swedish church (Heder Brandt et al., 2023).

All this work that involves the Swedish woodland, makes the industry responsible for between 9 and 12% of Sweden's jobs, exports, turnover and value (*Wealth creation*, 2024).

1.1.2 Harvesting wood

With the intention of managing forest resources, harvesting operations are required. These procedures play a significant role in obtaining materials and achieving healthy and sustainable forestry. Harvesting wood not only aims timber production but also ensures good management of wildlife habitat and reduces wildfire risks (Parajuli et al., 2020). It is considered one of the most cost-intensive operations when managing forests (Castro, 2016).

There are different ways of harvesting wood. One of the most used techniques is clear-cutting. In this case a whole section of wood is removed in one cutting, without leaving any left tree (Keenan, 1993).

Another used method is called thinning and consists of cutting only certain trees, removing just a selection from the whole forest. Thinning is thought to improve the health, growth and resilience of the left trees and it serves as a tool to decrease forest vulnerability (Moreau et al., 2022).

1.1.3 Terri

Here is where Terri plays a key role. Terri is a forestry machinery company that focuses on smaller machines. Their machines are smaller than usual, enabling them to harvest wood in tighter spaces. This makes them perfect for harvesting wood using thinning

techniques, which have a minimal impact on the soil and forests. Terri's forest machines consist of two parts. A cabin, where the driver takes control of the machine, and a trailer with a crane or a container depending on the type of machine (Figure 1). The inside of the cabin is the working space for the driver, in which they spend several hours daily (*Terri*, 2025).

Figure 1: Terri forestry machine



1.1.4 Forestry machine cabins

Working with smaller machines also implies a smaller cabin, and therefore a smaller working space. The working experience that a driver undergoes when inside a cabin is significantly influenced by the size and design of the cabin. Working in a cramped space affects the driver's comfort, and therefore their effectiveness. The layout of the cabin also plays a very important role, as it can significantly influence the operator's way of working and body posture. This issue is further supported by the fact that staying in the same position for extended periods is detrimental to health and can result in chronic fatigue and other health problems (Dvořák et al., 2020).

Operators in this field (Elliott et al., 2022) often work very long shifts, which can sometimes extend beyond 16 hours per day. During these long shifts, operators also perform daily activities inside the cabin, such as eating lunch or drinking coffee. As a result, belongings like lunchboxes need to be brought from home every day and have a dedicated space to be stored in the cabin. These are heavy objects and need to be transported repeatedly.

1.1.5 Ergonomics

Ergonomics aims to optimize the interaction between humans and systems. To reduce the risk of products' potential malfunctions, ergonomic principles should be considered during the design process. This way users' needs will be met, and issues such as accidents or health problems will be prevented (Soares & Rebelo, 2017). On the other

hand, products that perform their expected functions correctly, where ergonomics are considered, have less chances of leading to injuries and are more likely to be accepted (Karwowski et al., 2011).

Sitting in an improper position for long periods of time can lead to discomfort which can cause pain in different body parts: neck, shoulders, lower back, arms, wrists, legs and others. Over time, what at the beginning was only pain can lead to repetitive injuries (Sisay, 2016).

Therefore, having a comfortable and ergonomic working position can prevent future risks. Considering ergonomics in the automotive industry increases drivers' productivity and comfort simultaneously, since it improves driving conditions and satisfaction. Ergonomics plays an essential role in vehicle design from the initial phases of the process. Identifying the factors that affect the cabin's interior design and ergonomics will determine if the necessary measurements and needs meet the ergonomic requirements (Căşeriu & Blaga, 2023).

1.1.6 Previous research

In previous research carried out by students of the Enterprise course (including the author of this thesis) for Product Design within the master's Program in Product Design at Jönköping University during 2024, an investigation was conducted in collaboration with TERRI AB focusing on forest machinery. This study primarily addressed the interior and exterior design of the machines, resulting in improved redesign suggestions for both areas. However, while the analysis, some other aspects such as user related aspects were also considered, leading to the collection of insights about operators' needs, wants and pains points.

1.2. Problem Formulation

During the previous research done by the students in the Enterprise Course it was concluded that Terri's cabin can be considered cramped and lacks sufficient storage space.

Users spend long periods inside the cabin and often bring personal goods from home that they use in their daily life to make the working space more comfortable. A driver's comfort is essential for maintaining productivity. Current forest machinery solutions are trying to enhance comfortable environments for drivers. A well-designed working space, together with an organized storage system, ensures that the time spent searching for tools or equipment is reduced (Matteo, 2023).

Research by Product Design master's students observed that operators brought backpacks to the cabin to carry essential items from home that they needed during the

day. However, they did not have a dedicated space to place the backpack in the cabin, forcing them to place them on the floor, where they could obstruct movement.

1.3. Purpose and research questions

The goal of the thesis is to develop a bag that would solve the problem of forestry professionals' needing to transport everyday objects from their homes to their cabins. Additionally, this bag will be integrated into Terri's cabin design.

The research questions for this project are:

- *What are the existing challenges for forestry machine operators when carrying their belongings?*
- *What materials are better suited for a backpack design that prioritizes sustainability, meets users' needs, and withstands the conditions faced by forestry machine drivers?*
- *How to design a bag that supports effectively forestry machine operators during both work hours and off duty time?*

1.4. Delimitations

This final project focused on developing a design concept for a multifunctional bag, including the creation of a prototype and visualization resources. However, the project did not reach a deep level of industrialization. The cost and manufacturing aspects of the product have not been fully considered. The project centered around the company Terri and considered Swedish standards and regulations, without generalizing to other regions. The research will not go deep into ergonomics.

1.5. Disposition

The report is organized into 7 chapters.

1. **Introduction**. This chapter covers an introduction of the study, a brief background, purpose, research questions, and delimitations.
2. **Theoretical background**. This section consists of a compilation of the relevant knowledge to the field of research. It is used as a base to define the following methods.
3. **Methods**. In this chapter an explanation and description are presented of the chosen methods that were used during the product development process in the current project.
4. **Implementation and results of methods**. This section gathers a description of how the study has been carried out. It explains how the mentioned methods were approached and implemented and explains the results of each method.

5. **Final product results**. The final product results are presented to be able to answer the research questions and achieve the goal of the study.
6. **Conclusion and discussion**. This section included discussion of research questions and an analysis of the results given. This chapter will also draw a conclusion and suggestions on how the work can be further developed.
7. **Appendix**. This section gathers other material with additional information used along the project that may not fit in with the main body of the report.

2. Theoretical Background

2.1. Industrial design

Industrial design is the process of developing new concepts and products by taking specifications into account and focusing on different aspects. The first aspect that is considered is functionality, since products should be safe and intuitive. Form and appearance are also crucial in industrial design. Maintenance simplicity helps the product indicate how they should be conserved. Production costs must also be considered to ensure low costs. Reflecting the company's mission through the design will enhance the product's brand communication (Ulrich & Eppinger, 2012).

Industrial design is usually involved in different steps of the whole product development process. It is important that it ensures that products meet the needs of both users and manufacturers. These aspects lead to products or services giving the users a lasting value and overall experience (Ulrich & Eppinger, 2012).

2.2. Product Development

The product development process consists of a succession of steps that lead to ideate, design and commercialization of a product. It begins with identifying the market opportunity, then generating ideas and refining them, and specifying the product. The product must be reliable for production, for sale and for the final delivery. Usually, it involves screening down a broad number of concepts from the initial phases into a market-ready product. The manufacturing processes and the marketing plans also need to be defined. Shaping a structured development process is important to ensure product quality, teamwork coordination and good planning of the project. Strong collaboration across different areas is essential, especially during the early phases of the concept (Ulrich & Eppinger, 2012).

2.3. Research through design

As stated by Milton & Rodgers (2023), research is a structured process aimed at exploring new knowledge and information by analysing various sources and materials, which leads to new insights and conclusions. However, design research does not focus on what already exists, in contrast to scientific research, but rather on what it could be.

In context of research through design, with an experimental practice approach, research is contemplated as taking reference insights not related to the design field and adapted. This can involve an early idea or concept or discovering a process or new material. Research in this case is practice based and involves the creation of a domain and reflection on the process. The relationship between the designated products and the written description is what adds value to the research (Milton & Rodgers, 2023, pp. 11,12).

2.4. Human centered design

Human-centered design is a process that makes sure that the design identifies the correct problem and aligns with the needs and abilities of the people they are created for (Norman, 2013).

Human-centered design is an approach focused on having the user in mind and always considering its perspective in order to create effective and usable products, interfaces, services or systems. It adopts a problem-solving approach and focuses on the user's needs, desires, proprieties and capabilities, whether it's from a product, service or system. The aim is to improve the influence positively on the level of usability and the users' experience's quality. It helps predict future problems and aspects of the design. This type of approach is crucial when the gap between the designer and the user is big (Van Boeijen et al., 2014).

As stated by Van Boeijen (2014) , if the user is involved in the design process from the very beginning, it prevents to take important design decisions without considering the user's perspective.

2.5. Design thinking

Design thinking is a process focused on gaining a deeper understanding of users, questioning existing assumptions, reframing issues and developing innovative solutions to prototype and test. It is an iterative process where the goal is discovering alternative solutions and strategies that may not be immediately obvious based on the initial understanding (Dam & Teo, 2024).

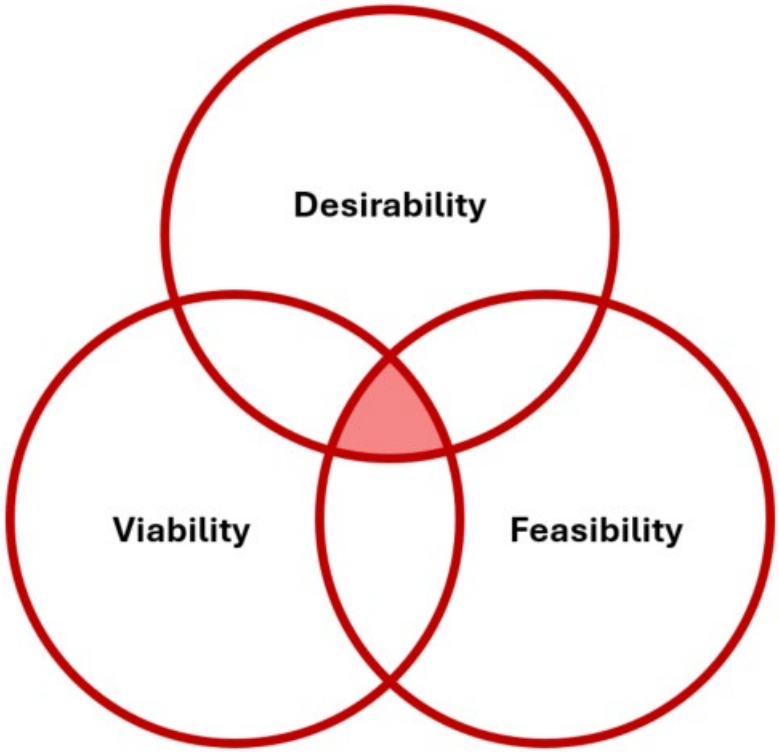
This human-centered process focusses on observing user needs, collaboration, quick learning, visualizing ideas and rapid prototyping. A designer's sensibility is typically applied, together with problem-solving methods, regardless of their nature (Lockwood, 2009).

According to Lockwood (2009), collaboration is a vital aspect of design thinking, includes both users and multidisciplinary teams. Involving the user in the early stages helps obtain evaluations of initial concepts too.

As part of the design thinking approach, designers must consider the DVF framework: desirability, feasibility, and viability (Figure 2). It is a strategic tool in product development that helps define challenges, rethink hypotheses and reformulate problems to enhance a creative ideation process. Design thinking guides multidisciplinary teams in evaluating concepts by considering the three critical aspects: desirability, feasibility and viability. It makes the design process more informed and fosters not only innovative design but also the development of sustainable designs that are viable in a real-world context. (O'Donoghue, 2023).

- Desirability refers to the alignment between the product and the user's wants. It concerns whether the users' preferences relate to the product, reflecting what they value and desire. Without desirability, a product lacks appeal and is less likely to succeed.
- Feasibility addresses whether the product can be developed from a technical approach. It involves analysing aspects such as resources, skills, technology or infrastructure to ensure the product can be created, as it must be realistically feasible to develop. If the product cannot be feasibly developed, even if it is appealing or potentially profitable, it will not be possible to produce.
- Viability relates to the financial aspects of the product. It must be sustainable over time and generate sufficient revenue to support its maintenance and development. Even when a product is desirable and feasible, without financial support it will not survive in the market (O'Donoghue, 2023).

Figure 2: Desirability, viability and feasibility graph



2.6. Design for emotion

According to Van Boeijen (2014), the design of emotions can play an important role in connecting a product to a person. Using emotions as a tool to achieve a deep connection increases the number of stimuli that the user has when interacting with a product. However, it is not only a response that the user receives during the interaction, but it also

reaches a deeper level. Humans can experience at least 25 distinct emotions when interacting with an object.

As well as objects, activities involved in the context of use can also be connected to emotions. The goal is that functional designs can become also impactful, memorable, and meaningful designs (Van Boeijen et al., 2014).

In align with (Van Gorp & Adams, 2012) designing for emotion goes beyond creating aesthetically pleasing products. Different aspects combine to communicate a product's personality, which is also crucial. This can create objects with a lasting impact and transforming objects into symbols. Our attention can also be affected by emotions. When many companies offer products that perform the same function with similar capabilities, users prioritize aesthetics and the emotions they feel when seeing the product.

In the design process, both cognition and emotion play a role. This means that great design should create positive and pleasurable experiences, such as satisfaction. However, there can also be overall negative experiences that result in confusion and frustration. But in order to obtain a successful product, the 3 levels of design should be considered: Visceral, behavioural and reflective (Norman, 2013).

This Design for Emotion can be used as an approach for 1) defining a suitable emotional effect, 2) collecting meaningful user information so that this emotional effect can be attained, 3) visualizing concepts that produce emotions and 4) measure how effectively the design concept evokes the intended emotion (Van Boeijen et al., 2014).

2.7. Semantics

As stated by Krippendorff (2006), product semantics is a detailed study of how people give meanings to objects and interact with them based on that meaning. It also includes vocabulary and methodology for designing artifacts while considering those meanings.

Those meanings will be interpreted by the users considering also the context of their experiences and interaction. It involves understanding both products' physical features and the cognitive and emotional ones that people associate with them. Krippendorff gives as an example the decision of choosing a car brand, since it can work to show a person's identity, status or values. The reason someone might buy an expensive car is not only for transportation purposes but to represent success or a certain lifestyle, which shows the social meanings connected to automotive brands (Krippendorf, 1989).

Semantics relates to the product sign's message. When applied to products, the "sign" is the way a product looks. The shape of a product can strongly indicate its purpose. For instance, smartphones all have the same basic rectangular shape in common (Sunde, 2018).

2.8. Semiotics

"*Semiotics is the study of sign and sign systems and their structure, proprieties and role in socio-cultural behaviour*." (Monö, 1997, p. 58)

Even though the sight is humans' dominant sense, when it comes to perceiving our surroundings, a person still uses the five senses to recognize the environment. Industrial design is therefore understood as something that should fit the whole person and their way of experiencing the surroundings through all their senses. According to product semiotics, meaning is attributed to what we perceive through our senses (Monö, 1997).

When applied to design, semiotic theories switch the products' meaning attention to the relationship between signs and what they represent, between signifiers and the signified (Krippendorff, 2006).

Semiotics is the study of signs and symbols within social contexts. Its purpose is to discover how the meaning is generated and communicated through different signs. Among those signs there are words, images, sounds and even products. A sign is believed to consist of two components. The signifier, which is the shape that the sign takes, and the signified, that is the represented concept (Ilstedt, 2002).

2.9. Materials experience

Material selection not only involves technological factors, since it will not be sufficient to fulfil every user's need. It is a combination of various factors what influences the final material decision, and therefore, makes it a multidisciplinary field. Material selection should also take the human aspect into account (Henriksson, 2021).

Human experience is formed through contact, meaning that the insights we gain through touch depend closely on the material proprieties that product has. As humans, we are instinctively inclined to see or touch objects, so that our senses have a reasonable explanation and a better understanding of our surroundings. We have developed the ability to recognize the characteristics or patterns in objects that are good for us and we tend to feel appreciation for them creating a materials experience for the user (Hekkert & Karana, 2014).

According to Hekkert & Karana (2014), in the same way a product can induce emotional responses, materials are capable of achieving them too. The unexpected strength or thickness of a material can be surprising for users, in a positive or negative way. Materials can be considered fascinating or disappointing to users and will affect products' perception. Consequently, it can result in either favourable or unfavourable emotional responses.

Materials driven design is a term mentioned by Hekkert & Karana, who emphasize the importance of introducing materials in the early phases of the design process. Karana

suggests not waiting until the concept is fully developed to determine the most suitable material but rather consider all the material options from the beginning. Observing material samples during the discovery phase can enhance the ideation process. Another approach includes selecting materials and investigating their design possibilities. Materials driven design does not aim for perfection from the beginning; by creating prototypes and handling physical materials, the goal is to foster creativity, rather than obtaining a finished result (Hekkert & Karana, 2014).

2.10. Design for Sustainability

Design for Sustainability (DfS) is a design approach that aims to have a beneficial impact on the environment and society. Creating any new product or product-service system will have an influence, which is why designers should always keep sustainability in mind, as it is a responsibility that must be considered. It is fundamental for both humans and the planet's well-being. Doing sustainable design means identifying which factors are important when developing new products so that new solutions with long-lasting consequences are created (Van Boeijen et al., 2014).

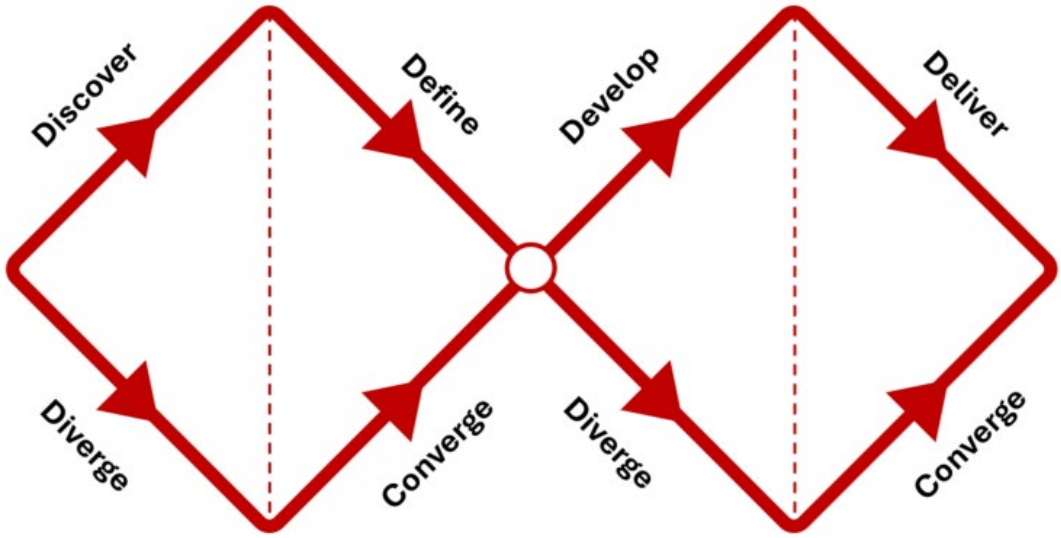
One very important factor to consider in design for sustainability is transparency, both in the manufacturing and in the materials used. Traceability is a key factor in manufacturing and supports the development of a sustainable supply chain of the resources needed. Greater visibility into environmental impact and production processes benefits both the company and outside stakeholders, by increasing trust and ensuring they follow regulations (Santharm & Ramanathan, 2022).

3. Methods

3.1. Double diamond design process

The Double Diamond is a design framework which focusses on exploring a design problem more widely, divergent thinking before taking targeted action (convergent thinking) as shown in Figure 3. It was firstly introduced by the British Design Council in 2004 (British Design Council, 2025).

Figure 3: Double Diamond design process



The double process of divergence and convergence releases designers effectively from unnecessary limitations to the problem and solution. This iterative approach of diverge-converge is essential to correctly design the correct problem and which way it needs to be solved. (Norman, 2013). It is divided into 4 stages:

- **Discover.** The first diamond starts by researching and identifying what are the problem and the user needs. It consists of understanding and spending time with the people affected by the issues.
- **Define.** The second step involves the information obtained during the discovery phase, which will be analysed to define the problem correctly and make sure that it aligns with the user's needs.
- **Develop.** The second diamond takes the clearly defined problem and focuses on developing, creating and testing multiple potential solutions.
- **Deliver.** The final phase includes choosing the best solution, testing it and getting it ready for launch.

3.2. Discover

The discovery phase is the first step of the double diamond framework, and it begins with an exploration to determine what the main problem and the needs of the users are. It involves gaining understanding and dedicating time to the users who are impacted by the problem (British Design Council, 2025).

3.2.1. Competitor product analysis

A competitor product analysis can help to develop a clear picture of what kind of products exist in the market. This method consists of investigating products that address similar problems in the sector. The products are then evaluated to find the advantages and disadvantages of each of them, which will bring helpful insights into the new design. The data collected can be of different kinds. Quantitative information is useful for comparing functions and requirements. But qualitative knowledge or more subjective data is also collected. A selected number of competitors are listed in which different aspects will be compared. The next step is summarizing key findings and areas that should be considered and solved in order to compete effectively in the market (Milton & Rodgers, 2023).

3.2.2. Materials research

Materials research is part of the design development process. In product design a material analysis refers to an investigation among the materials possibilities in order to select the most suitable materials, in which various factors and constraints need to be considered. In most cases, product requirements can be fulfilled by more than one material, and the final choice has to consider the benefits and trade-offs of each option (Karana et al., 2008).

Choosing the correct material is essential in successful product development processes. Materials research consists of selecting, analysing, and comparing different materials relevant to the design. Utilizing this tool can help designers to understand which constraints to consider. Characteristics such as durability, price, sustainability, waterproofing and aesthetics, among others, are relevant to decide which option is the most suitable (Sun et al., 2023).

3.2.3. Questionnaire

A questionnaire is a research tool that can be used to gather information by asking individuals a list of questions. It is classified as a quantitative data collection technique since it can collect large amounts of information and can, therefore, be helpful when a numerous amount of people is involved. Designers use questionnaires to get to know users' opinions or preferences in various fields as well as their motivation for choosing between different products (Van Boeijen et al., 2014).

This tool is useful during the first phases of the project considering that it can help identify the most relevant target group. However, questionnaires can also be valuable in later stages for evaluation concepts and deciding between multiple concepts as it can indicate users' acceptance of a concept (Van Boeijen et al., 2014).

3.3. Define

The second stage of the double diamond is Define. It focuses on analysing the information gathered during the previous phase, so that the problem can be accurately identified. This ensures that the issue is aligned with the users' needs (British Design Council, 2025).

3.3.1. Persona

The Persona method aims to define an archetype, an imaginary person that fits the typical characteristics of the intended user. It is particularly helpful when segmenting groups of users, especially if there are different types of users. The goal of the persona method is to list detailed information that would describe the user deeply, such as the name, age, education, family and friends, lifestyle (including also the type of brands or products they consume) and hobbies. It provides designers with a useful tool to get to know better who a real user might be. By considering a stereotypical user as the main objective, it leads to fulfilling the needs of a broader group of people (Milton & Rodgers, 2023).

According to Van Boeijen (2014) to create the persona, already existing customer data must be used to gather insights about their goals and behaviours. This information is collected during the exploratory phase. Methods like interviews or observation are effective for collecting qualitative data necessary to develop the persona.

Persona is a straightforward approach to getting closer to the user and making meaningful decisions. Having a clear idea of the purpose and forming a shared image of the users' values will help later during the design process. The defined persona will be used during the whole product design development process (Van Boeijen et al., 2014).

3.3.2. Empathy map

An empathy map is a design tool that symbolizes what designers understand about the user and develops a clear picture of how they actually feel fostering a more human centered mindset. By collecting qualitative data considering their feelings, experiences and thoughts and analysing them, it can give a better picture of their needs and help comprehend their emotional responses (Lammers, 2021).

Typically, empathy maps include various sections:

1. What the user says: listing quotes or feedback the user has said.

2. What the user thinks: internal thoughts that are usually not shared.
3. What the user does: actions and usage of the product.
4. How the user feels: emotions and feelings the user gets when utilizing the product.

By collecting this information, designers can take more thoughtful decisions that are more empathic and create solutions that respond to their challenges and products genuinely (Gibbons, 2018).

3.3.3. Journey mapping

Journey mapping consists of visualizing all the steps in which the user interacts with a product or service and can be understood both in terms of time and place. Understanding people's experiences can lead to new solutions for each step of the process, since it consists of picturing what the user would like to use. This map is based on a specific type of user, the user persona for the project, and focuses on the emotions and behaviours they go through. It is divided into various phases each representing a step or action the user must take when interacting with the product (Van Boeijen et al., 2014).

For designers, this method helps identify the pain points as well as the emotions that a product could generate in the user. Journey mapping is based on qualitative data that can be collected from previous methods and is not based on designers' assumptions. The horizontal axis represents time and a vertical axis, which is used to list the themes and concerns related to each project. After exploring the journey map, opportunities can be identified to improve both small actions within the interaction with the product and the overall user experience (Van Boeijen et al., 2014).

3.3.4. Workshop

A workshop is a qualitative research method aligned the human-centered design approach. It is a structured and organized session in which a group of people collaborates to discuss, explore or solve specific design problems. There is usually someone with expertise in the topic that guides the session. However, while the workshop has a defined goal, its outcome may not be fully predictable at the beginning. Workshops in design include collaborative activities such as generating ideas, with a strong focus on the interaction between individuals like designing concepts or prototyping (Ørngreen & Levinsen, 2017).

Active participation can lead not only to new solutions but also to changing the direction of the workshop. This helps translate abstract ideas to a final outcome, which normally aims to be something tangible such as concepts, solutions or prototypes. Usually, workshops consist of a small group of people so that they reach a deeper level of reflection. The participants are often different stakeholders to provide diverse perspectives on the problem addressed (Ørngreen & Levinsen, 2017).

3.3.5. User needs identification

To gain a comprehensive understanding of users and their concerns, it is essential to identify their needs. The user's needs identification is a process that might change from one project to another, but in most cases consists of five steps. First, a collection of quantitative and qualitative data from different sources like questionnaires, interviews or workshops takes place. Once the information is gathered, it is analysed and interpreted and later collected in a customer´s statements list. Customer statements are opinions that users have expressed and were obtained through the selected data collection techniques. These statements are then organized into a hierarchy of primary, secondary or tertiary and then considering their relative importance between each other as must-have, should-have or nice-to-have. Finally, after the analysis, conclusions about the results are synthesized and the customer statements that were initially phrases are translated into defined user needs (Ulrich & Eppinger, 2012, p. 75).

It is important to remember that customers' perceptions and preferences, such as product's appearance, aesthetics, affection, usability or comfort, are typically classified as subjective, yet they should still be contemplated as valid user needs. It encourages a more human-centered approach, which often is forgotten in more traditional technical design practices. However, subjective qualities like emotional responses, aesthetic or user friendly design are just as equally relevant as technical aspects (Yue Wang and Mitchell, 2011).

3.3.6. Product list of requirements

The "language of the customer" is the way customer needs are typically expressed. This step is very useful for gaining proper understanding of what the user is trying to communicate as their needs. However, this kind of "language" provides vague information that may be open to subjective interpretation, as it does not offer clear instructions for designing the product. To address this issue, product development teams must translate user needs into product requirements which the product must fulfil. These requirements do not specify how the design needs to be addressed, but instead define the goals the product must accomplish to satisfy the user's needs (Ulrich & Eppinger, 2012, p. 92).

After listing the customers' statements and identifying the user's needs, the designers' task is to translate them into engineering product requirements that also include specific technical information. These requirements should reflect both technical aspects, as well as emotional considerations to ensure that the product criteria are fully defined. Effectively translating what the customer's wants and needs into valuable requirements is dependent on strong communication between teams involved in product development (Michael, 2009, p. 24).

3.4. Develop

When the first diamond of the double diamond design process is completed by defining a clear problem, the second diamond builds on that foundation. The focus then shifts to generating, developing and testing multiple potential ideas to solve the problem (British Design Council, 2025).

3.4.1. Brainstorming

Brainstorming is a creative technique that encourages the free generation of ideas and fosters innovative thinking without any restrictions. It is commonly used in group settings, stimulating collaboration, but it can also be used as an individual method. The goal is to stimulate creative ideation and obtain as many ideas as possible. It is particularly effective in nonjudgmental environments (Wilson, 2010).

This method is based on different core principles. As previously mentioned, it enhances the generation of a high number of solutions and the ideas cannot be judged until the end of the session. Unconventional ideas are encouraged. The combination of existing ideas can lead to interesting combinations (Bonnardel & Didier, 2020).

In addition to brainstorming, sketching can be used as a tool to picture ideas in a physical format. As Lugt mentions, “thinking sketches” are drawings made by designers to support their personal ideation process. Using thinking sketches can facilitate visual reasoning (Lugt, 2005).

3.4.2. Try it yourself

To develop a clearer perspective of the user’s actual experience, the method ‘Try it yourself’ is typically used. This research method helps designers increase their awareness about how users interact with a product or service. This information gathered by hands-on exploration can be a way of inspiring ideas in new directions while keeping user needs in focus. ‘Try it yourself’ requires designers to use the real product in real-world conditions and not in a controlled environment in a studio. Taking notes and pictures of the process is very valuable for identifying challenges and reflecting on initial assumptions (Milton & Rodgers, 2023).

3.4.3. Concept generation

The concept generation method involves developing multiple design ideas that are defined as concepts. A product concept refers to an initial outline of a product’s functionality, technology and form. However, it offers a clear description of the way the design will meet the user’s needs. Typically, this concept is represented with tools such as sketching or rough 3D models and are complemented with a textual description of it. The process of concept generation begins with identifying the user’s needs and target

goals and lead to identifying a range of concepts from which the final product concept will be selected (Ulrich & Eppinger, 2012).

Concept generation involves a complex process in which decisions about the different aspects need to be addressed. The customers' needs are considered but the designer's perspective must be acknowledged too. Among the considerations, functional requirements and interaction between components are equally important to consider. (Fazeli & Peng, 2022).

3.4.4. Mock-ups

Creating a mock-up consists of developing from simple materials like cardboard product concepts. They allow designers to visualize simplified functions, shapes or ergonomics of a product, without having to recreate a fully detailed version. Mock-ups also assist in the physical evaluation and interaction by testing surface features such as colour, texture or specific details and show better insights of the proportions of the design concepts, that are usually difficult to judge through sketches or technical drawings (Milton & Rodgers, 2023).

Unlike computer modelling or simulations that are based on assumptions and approximations of reality and might not relate completely to actual product behaviour. On the other hand, mock-ups offer a tangible way of interacting with the product and testing its performance. They are valuable to prove the feasibility and validate design options, as well as identify areas for improvement (Milton & Rodgers, 2023).

Mock-ups are typically produced using inexpensive resources that mostly focus on functionality rather than final styling. Although the number of mock-ups can vary depending on the project and budget, their low production cost can prevent costly mistakes (Milton & Rodgers, 2023).

3.4.5. Concept evaluation and selection

Concept evaluation plays a crucial role in the decision about which of the different design alternatives best meets the established requirements. This process represents a multiple-criteria decision-making process (Fazeli & Peng, 2022).

The process of concept selection evaluates concepts based on customer needs and other criteria. The different ideas are compared to one another focusing on their advantages and disadvantages. Afterward, the selected concepts will be further investigated, developed and tested. Concept selection is one of the steps in the product development process. While in previous steps the focus was diverging thinking and creativity, when selecting and screening the concepts designers must use a more convergent approach to reduce the number of options. It is, however, an iterative process, since it can take place more than once throughout the whole design process.

Even if the goal is to narrow down the number of concepts, on some occasions concepts might be combined and redefined which can lead to an increase in the number of concepts again before taking the final decision. Concept selection can be done in many different ways. In some situations, using an evaluation matrix is the most suitable, although in others using pros and cons or listing the advantages and disadvantages can be more appropriate. Another technique can involve developing a prototype of each concept and testing them, then making a decision based on the results (Ulrich & Eppinger, 2012).

3.4.6. User concept testing

Concept testing is a stage of the product development process that allows designers to collect direct feedback from potential customers about a specific product. In this phase, the design team describes a concept to the target group and gathers their opinions and insights. The purpose of this method is to gather perspectives from potential customers that will help decide which concept should be selected. Valuable insights about redefining ideas or improving the concepts can also be gathered, which improves the potential success of the product in the market (Ulrich & Eppinger, 2012).

Concept testing typically follows concept selection since testing a large number of concepts with customers can be difficult and not feasible. Additionally, concept testing is closely linked with prototyping as the presented concept is often accompanied by a prototype to help the customer understand it and provide more accurate feedback. For the concept testing process to be successful, it is recommended to prepare a list of questions in advance. Having a clear understanding of the test's purpose is essential for the development team to achieve effective insights. By establishing specific goals, designers can systematically repeat the test with different customers and develop a method that aligns with the desired outcomes. A group of possible users is assembled to participate in the tests. The potential customers will have to answer questions about the intended product. The next step is to choose the format for the survey to be conducted. Among the different possibilities that exist, some could be face to face, via telephone, email or internet. With the results, the design team can make a more informed decision about which concept is more appealing for the users (Ulrich & Eppinger, 2012).

3.5. Deliver

The final phase of the process involves selecting the most effective solutions. Testing and refining must be completed before selected solution is ready for the final launch (British Design Council, 2025).

3.5.1. Prototyping

Physical prototypes are tangible artifacts that serve as tools to explore and test a product's characteristics. These tangible models can range from visual mock-ups to experimental hardware to showcase functionality. However, analytical prototypes are intangible representations, that can be simulations or 3D models (Ulrich & Eppinger, 2012).

Prototypes can also be divided into comprehensive or focused types. Comprehensive prototypes represent a fully operational model that relate closely to the final product. It is used to find any final flaws. On the other hand, focused prototypes are limited to only exploring specific features in depth. They can be, for example, cardboard mock-ups, foam models, or basic circuits for performance testing. A common practice is to use both types of prototypes. One that is similar to the final product aesthetic-wise, but that doesn't relate technically, and the second one only focuses on the functions. This strategy allows a more efficient evaluation of the product instead of building a high-fidelity prototype (Ulrich & Eppinger, 2012).

4. Implementation, results and interpretation of methods

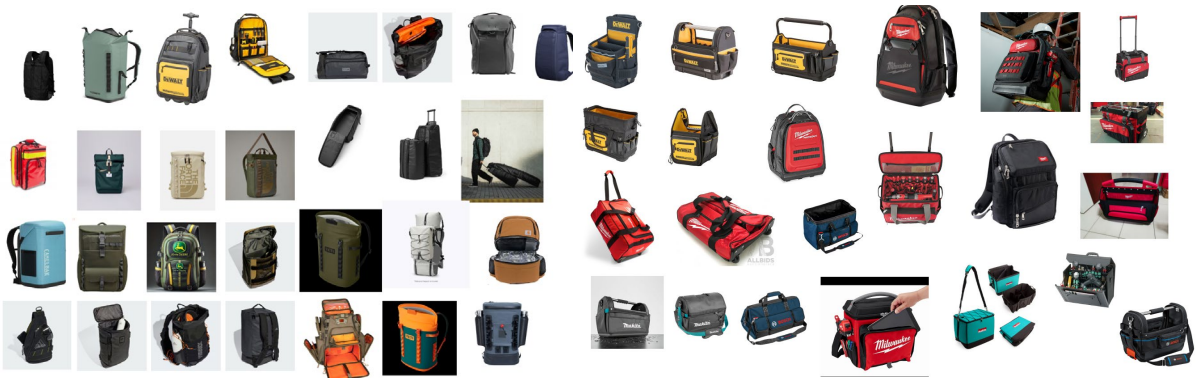
This chapter applies the methods within previously discussed Double Diamon Design framework. It details the selected approach, describes the implementation process and presents the results obtained from each method.

4.1. Discover

4.1.1. Competitor product analysis

As part of the competitor product analysis, a study of the current backpack market was conducted to gain valuable insights into existing products. The selected approach was searching for bags available on the market that had at least one feature that could be useful for carrying belongings for forestry machine operators. First, several bags were examined to find the most suitable types and their features were analysed. The main types of products considered included backpacks, toolboxes, lunch bags and camera bags (Figure 4).

Figure 4: Competitor product analysis



For each different kind of bag some conclusions were made. Backpacks were the first product analysed as they are a popular way of transporting items. They can generally be divided into two main groups: everyday backpacks and technical backpacks, such as those designed for hiking, sports or transporting tools. Across both categories, bags often had common features, including outside pockets in addition to the main one, padded backs and foam inserts in the back side to provide a more comfortable fit. Some bag designs had wide-opening zippers, providing easy reachable pockets. A key aspect considered during the analysis was storage capacity, in which maximizing volume was a priority. This resulted in square or boxy shapes in most cases. Modularity and multiple compartments were also common characteristics found in many designs.

Toolboxes offered insights into how to implement different compartments for various purposes considering a wide range of tools. Most were developed by a tool company thought specifically for carrying their equipment. Compartment arrangement in toolboxes provided designated sections for specific tools, as well as more adaptable

storage for general use. Large handles were a recurring element to ensure even weight distribution and ease of carrying. To have good stability toolboxes tend to have flat and rigid bases and hard walls, with a strong emphasis on straight edges and corners.

Lunch bags were generally square-shaped, since they were designed to accommodate food containers that typically share that form. Insulated foam lining within the walls helped maintain food temperature and they often included a waterproof fabric cover to allow easy cleaning in case of leaks or spills. When multiple compartments were included, secondary pockets were usually designated for smaller items such as cutlery or napkins. For transportation, lunch bags featured two small handles to carry around with one hand and an additional shoulder strap.

Camera bags on the other hand, were characterized by their high number of different compartments. Their main consideration was the customizable pockets, which could be adapted thanks to Velcro strips along the sides and internal walls to adjust the side of the pockets to fit different objects. They also were equipped with shoulder or back straps to allow hands-free carrying.

After analysing these various types of bags, the analysis indicated that a backpack was the most suitable solution for the design challenge addressed in this thesis. However, valuable features from the other bag types were also be considered for the concept generation. Key insights such as high compartmentalization, leak protection and modular pocket systems have proven to be particularly useful to improve functionality in this bag design.

4.1.2. Materials research

The material research in this project began at the very start of the design process, following a materials-driven design approach. This was crucial not only to satisfy technical requirements later in the development process, but also to consider the overall user experience of the final product, which will be an outdoor backpack.

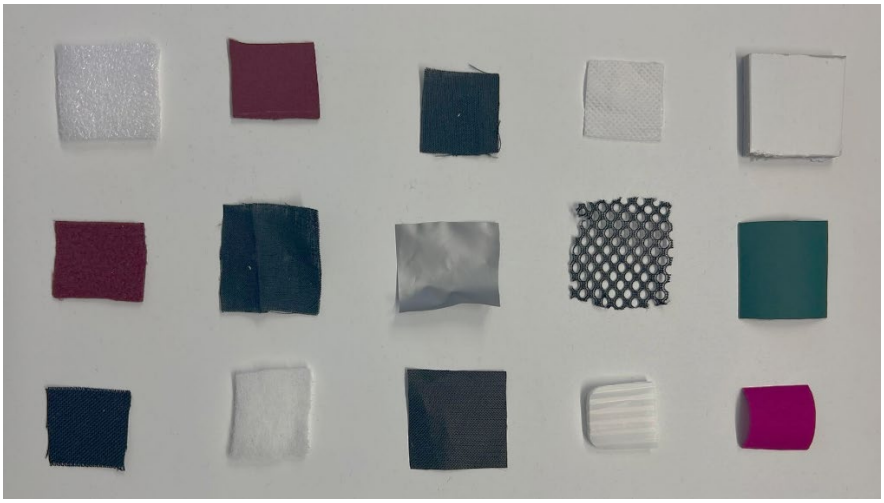
The research consisted of analysing a list of materials widely used for backpacks designs. Some of the aspects considered were their functionality, sustainable aspects, as well as emotional responses. Materials like ballistic nylon, cordura, polyester (PET), TPU-coated fabric, waxed canvas, ripstop, and also some recycled options such as recycled polyester and recycled nylon were evaluated. Each material was rated considering the following criteria: durability, weight, weather resistance, cost, and environmental impact. After that, a comparison table was created where the advantages, disadvantages and ideal applications of each fabric were summarized.

However, material selection was not only guided by technical performance. As Hekkert & Karana (2014) suggest, materials can influence the emotional connection between a product and an individual, and therefore shape users perception and interaction with the

product. For this reason, tactile exploration played a key role. Physical samples were examined and tested to observe their feel, texture, weight or flexibility. Quality perception was also a key point to analyses, because perception of comfort and functionality can be dependent on that too. This approach reflects the view that material selection is multidisciplinary, where designers have to combine technological, aesthetic, and emotional factors (Henriksson, 2021).

This material board shows various samples of materials, each of which was tested observing their texture, appearance and performance to provide material insights for the designer (Figure 5).

Figure 5: Materials board



Following the material analysis, recycled polyester (rPET) was selected as the most suitable material based on its sustainability, balanced technical performance, and appealing tactile qualities that transmitted strength and that are aligned with the demands and conditions the bag design will have to endure. For a more detailed description of the analysis chapter, refer to the appendix chapter.

For the inside lining of backpacks there are also several options that serve similar purposes. Backpacks usually use different types of foams to ensure the users comfort, to give structure to the bag and enhance durability. Among the different possibilities there is PE foam, which is firm and has water resistant properties. The use of PE is mostly for either structural use or outdoor use. Another material is PU foam, which is a soft material usually used for padding, in backpacks for example. EVA foam has shock absorption properties, and it is mostly used to protect electronics from damage. Memory foam provides ergonomic comfort to the product. It exists also some recycled foams that can be used in some designs. PE was chosen a the selected one for padding, and it is a petroleum-based material, but if it is obtained from recycled sources, such as industrial waste or repurposed packaging, it can help minimize its environmental impact.

4.1.3. Questionnaire

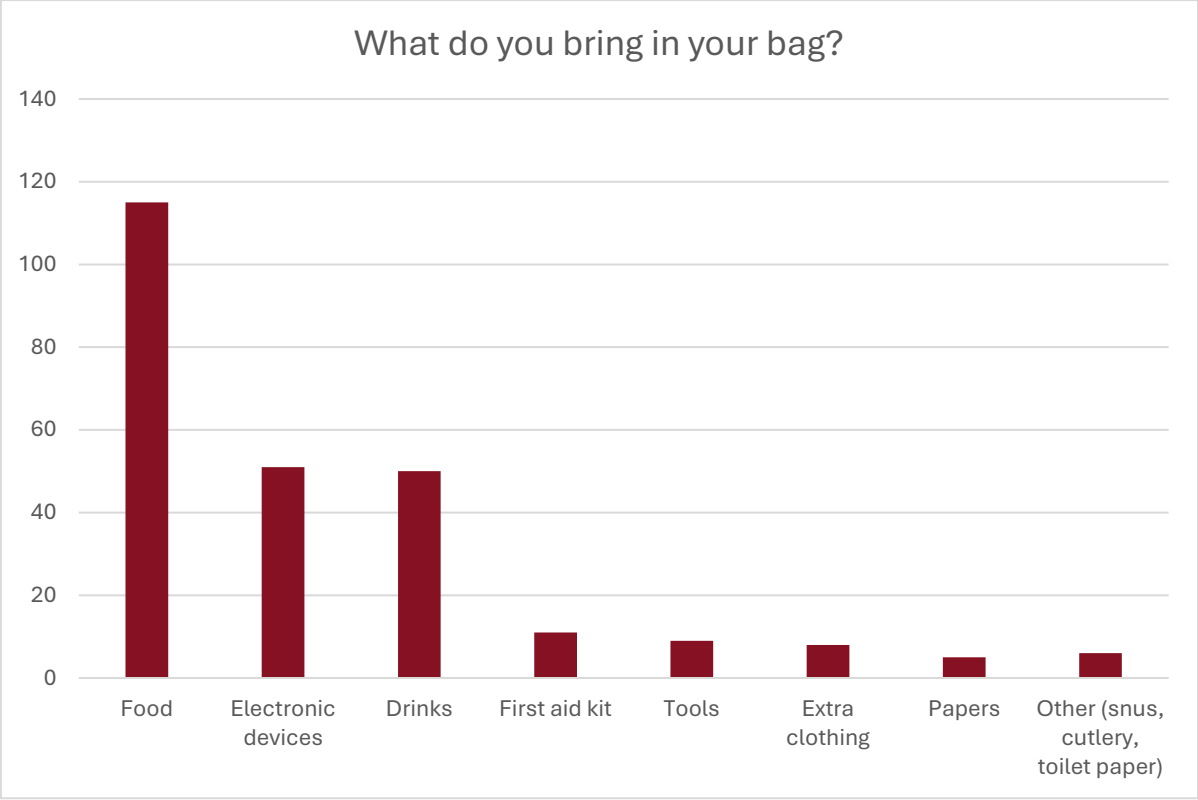
To support this investigation, a questionnaire was distributed to gather information about users' design preferences and their daily routine. Based on the responses, a preliminary user persona was developed, providing valuable understanding of the target group.

To examine more thoroughly the daily routine of forestry machine operators, a targeted questionnaire was designed and distributed to the relevant group. Preliminary research was carried out to identify the most appropriate type of questions. Particular attention was given to the earlier study conducted by the Enterprise course in Product Design students, which provided useful insights into the users. The final questionnaire was shared in two different Facebook groups dedicated to forestry machinery in Sweden, ensuring direct access to a relevant and engaged audience.

The survey included general questions regarding age and machine usage and were followed by more specific questions concerning operators' transportation of their personal belongings. Respondents were asked about how they carry their items, opinions on the existing solutions they have, and what improvements they wish to have. A total of 135 responses were collected, offering valuable data for further analysis.

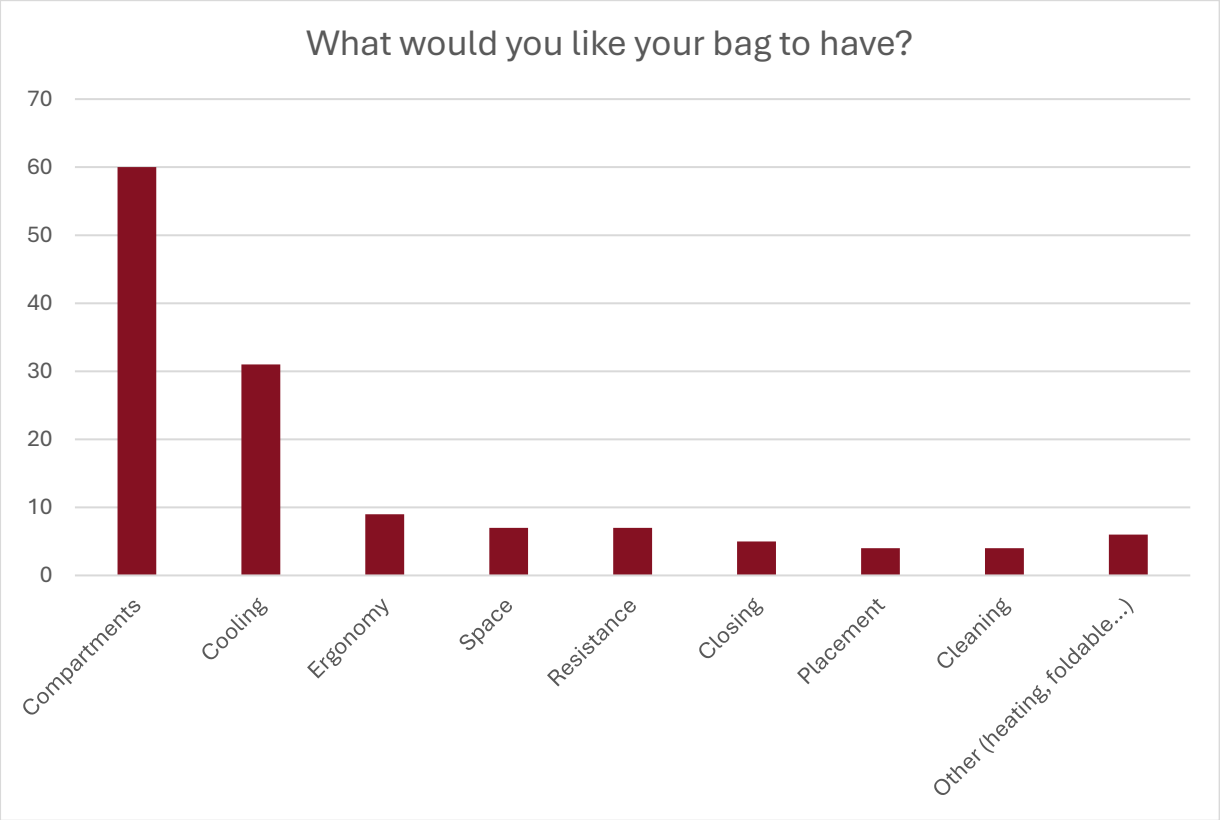
The results revealed that forest machine drivers often need to walk with their bags shorter distances than was initially expected. Typically, they carry items such as food, cool drinks, and electronic devices (Table 1).

Table 1: Questionnaire answer: bag items



Sometimes, they bring an extra lunchbox or snacks to complement the one they use for lunch. The most valued features in a bag were the presence of multiple compartments, ergonomic carrying, and the ability to store the bag securely in a designated space within the machine (Table 2). Stability was emphasized, since users expressed the need for a bag that remains straight and prevents contents from falling while the machine is in operation. Overall, convenient and functional design was identified as key user priority.

Table 2: Questionnaire answer: Users' wishes



For a more detailed description of the questionnaire answers, refer to the appendix chapter.

4.2. Define

4.2.1. Persona

To better understand the forestry machine operator user, who represents the main user of the backpack, the user persona method was conducted. This method allows designers to get to know the user's needs, behaviours and preferences. Based on the results of the questionnaire and the previous research conducted by the Enterprise course students, a detailed user persona was developed to guide the design process.

Erik Olsson is the user persona TERRI's backpack. He is 34 years old and works as a wood harvester with a TERRI forest machine. He is from Halland, Sweden and has over 15 years

of experience in the forestry field and considers that his work is closely aligned with his personal interests. Anders is skilled in his job, although he did go to university.

Erik is open to innovation, but he uses technology moderately and it might require him some time to adapt to unfamiliar systems. Functionality and order are key points for him, but he must work in a cramped machine cabin that does not have much space and lacks organization due to the limited number of compartments inside the cabin. He usually works long shifts in the forest as part of his daily routine, which starts at five in the morning. Family is very important to him in his personal life. Erik really enjoys outdoor activities such as motorsports and has a keen interest in mechanical engineering topics which he often consumes. Erik needs a functional, ergonomic working space with storage that is intuitive, as well as multiple compartments to organize his tools and belongings.

4.2.2. Empathy map

An empathy map was created to better understand the users' insights collected from the questionnaire. Some of the responses were subjective and could be interpreted in different ways depending on the context. Developing a better understanding of how users feel enables designers to create a product that completely fulfils the user's wants. Along the empathy map, different aspects were considered: compartments, stability, placement, ergonomics, closing, cleaning and cooling. The map was divided between what the users say, what the users think, what they do and how they feel. Statements were collected from the questionnaire and feedback provided by actual forest machine operators. Their main pains gathered had to do with issues they experience during their working day (Figure 6).

This analysis helped conclude that accessing the backpack from inside the cabin can be difficult and can lead to the backpack moving and resulting in items falling. Consequently, they have a cramped and unorganized cab which is their working space. When users described what they wanted a convenient backpack was that it works well and does not develop these pains in their experience using the product. The product should avoid the following issues: remain stable, is comfortable to use, can be closed properly and has many compartments. The backpack should also have a cooling feature and be easy to clean. Lastly, it will be convenient to have a designated space for the backpack within the cabin so that it does not move around the floor of the cabin.

This information was validated by the insights obtained from the questionnaire and the feedback and pictures sent by actual users.

Figure 6: Empathy map



4.2.3. Product Journey mapping

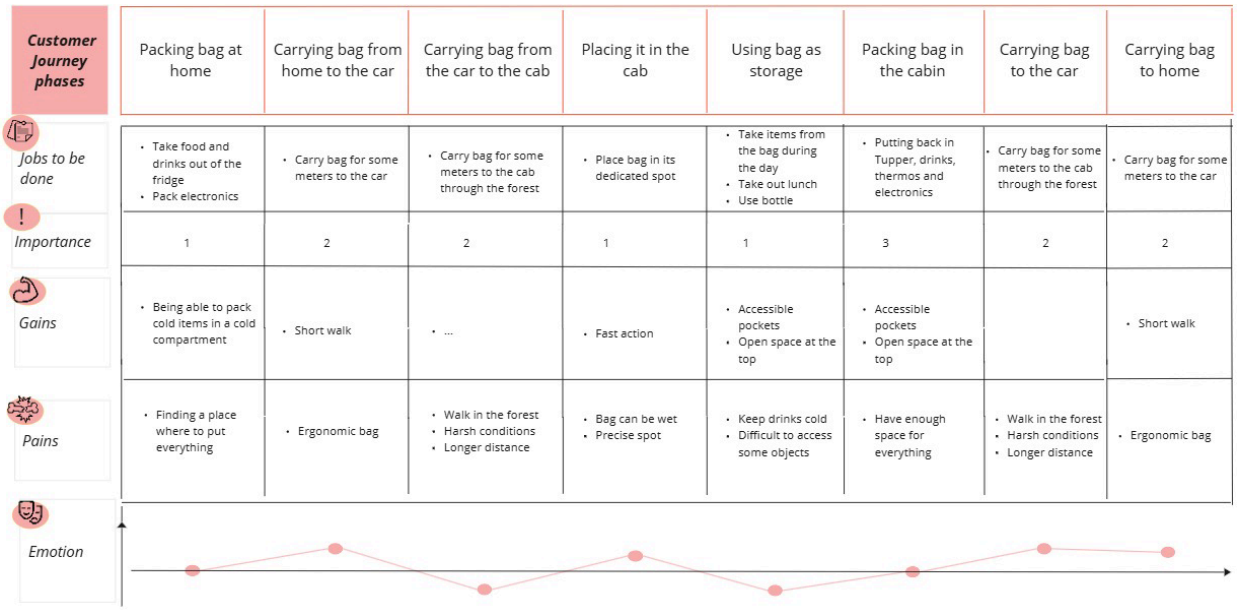
From the qualitative data gathered through user feedback and from observation conducted in previous research by students from the Enterprise in Product design course, a journey mapping was developed. The map was not based on designers' assumptions, but on real user experiences (Figure 7).

This customer journey of the backpack explores each step of interaction between the user and the backpack and was structured into eight different key steps. The first step begins at home, where the operator packs in his bag everything he will need during the day, like food and drinks. This task can be hard if there is not sufficient space in the bag or is a lack of organization. After that, the user must carry the backpack from home to the car, which usually is a short and manageable distance that does not require much effort.

Once the operator has arrived at his destination with the car, he must take the bag and transport it from the vehicle to the cabin, which depends on how far away the machine is, how inside in the forest it is. From the data collected in the questionnaire, although it can be assumed that the distance is usually not long, the process can be unpleasant in harsh weather conditions such as rain or snow. Once inside the cab, the bag is placed inside for the rest of the workday. During their shift, operators often need to access items from the bag. However, many reported frustrations during this step since it is difficult and annoying to reach their belongings comfortably from the bag. At the end of the day, the same sequence of actions is repeated in reverse.

Analysing the journey mapping revealed two main pain points: carrying the bag from the vehicle to the machine through the forest in rough conditions and accessing the contents inside easily while working in the cramped cabin space.

Figure 7: Product Journey mapping



Customer Journey phases	Packing bag at home	Carrying bag from home to the car	Carrying bag from the car to the cab	Placing it in the cab	Using bag as storage	Packing bag in the cabin	Carrying bag to the car	Carrying bag to home
Jobs to be done	• Take food and drinks out of the fridge • Pack electronics	• Carry bag for some meters to the car	• Carry bag for some meters to the cab through the forest	• Place bag in its dedicated spot	• Take items from the bag during the day • Take out lunch • Use bottle	• Putting back in Tupper, drinks, thermos and electronics	• Carry bag for some meters to the cab through the forest	• Carry bag for some meters to the car
Importance	1	2	2	1	1	3	2	2
Gains	• Being able to pack cold items in a cold compartment	• Short walk	• ...	• Fast action	• Accessible pockets • Open space at the top	• Accessible pockets • Open space at the top		• Short walk
Pains	• Finding a place where to put everything	• Ergonomic bag	• Walk in the forest • Harsh conditions • Longer distance	• Bag can be wet • Precise spot	• Keep drinks cold • Difficult to access some objects	• Have enough space for everything	• Walk in the forest • Harsh conditions • Longer distance	• Ergonomic bag
Emotion								

4.2.4. Workshop

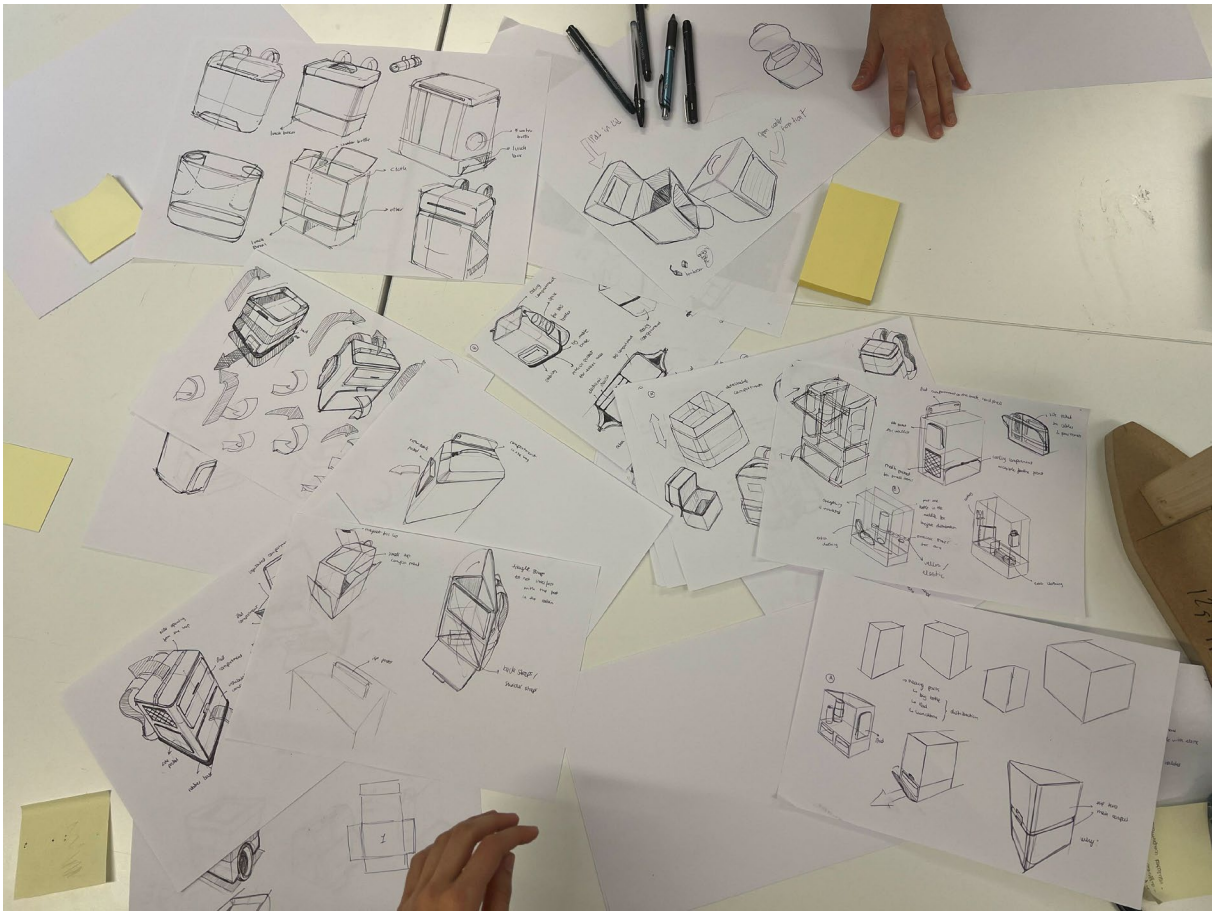
During the sketching workshop, the shape and arrangement of the compartments were discussed. The workshop involved four participants, including design classmates and the thesis supervisor, and lasted 1 hour. The session was divided into three parts. First, a sketching phase that lasted 15 minutes, followed by a 15-minute presentation phase of explanation and analysis of the sketches and finished with a 30-minute group discussion (Figure 8).

The most important topics addressed were weight distribution and, in relation to that, the placement of the different compartments. During the first part, each person tried to come up with innovative ways of organizing the different compartments considering the three following aspects: creating a multifunctional backpack, including numerous compartments, and defining a place for food and drinks. After the sketching session, a discussion phase followed. Each ideation process was explained and discussed to identify the most valuable points. Finally, another 30-minute discussion was held to further explore potential solutions for dividing the backpack.

The results from the workshop indicate that the water bottle should be placed on one side for easy accessibility from the outside. To achieve a good weight distribution in the bag, the two lunchboxes should be placed on one side, stacked on top of one another. This arrangement ensures that the two heaviest items in the bag will be at the bottom with one item on each side for balanced weight distribution. Regarding the shape, it was

concluded that the square shape is the most efficient to utilize all the space when in the cabin, since the placement of the bag in the cab will be in a corner.

Figure 8: Workshop sketches



4.2.5. User needs identification

Through various data collection methods, a significant amount of information was collected. This data needed to be carefully analysed in order to be considered to inform the design process. From the questionnaire responses and direct user feedback, a list of customer statements was identified. These statements reflect users' thoughts and are mostly related to the challenges they currently experience when carrying their personal belongings. The key mentioned topics included stability, ergonomics, closure mechanism, compartmentalization, difficulty of cleaning, and cooling features. There were mostly difficulties reported when using their own bags, especially when inside the cabin.

After analysing and summarizing the customer statements, they were translated into clearly defined user needs. Users need a backpack that is easy to carry, allowing them to keep their hands free for other tasks like holding a flashlight. They need to carry essential items that they will use during their shifts like food, cold drinks, electronic devices or other objects. Since their working environment is usually tight and cramped, they want a

dedicated fixed place to store their bag is essential to prevent it from collapsing during their working shift. Furthermore, operators desire a backpack that allows easy access to its contents and keeps items organized and in place. When bringing food, users emphasize they would like to prevent any spill or leak, and if these occur, to be able to clean the bag easily. Given their long shifts in both cold and hot seasons, they want to maintain cool temperatures for drinks.

For a more detailed description of the user's needs and customer statements, refer to the appendix chapter.

4.2.6. Product requirement list

The requirements list works as a bridge between the user needs and the conceptual design solutions. Once the user's needs were identified, they were transformed into specific product requirements, to make sure the users' expectations and preferences are aligned with the final product in real-world conditions.

The requirements list includes functional, ergonomic and environmental considerations. Some of the requirements are that the backpack must stay upright and be stable on its own. It must be easy to open and close, even when wearing gloves or under harsh conditions, and when closed it should provide a secure closing. The bag should include back straps and handles. These features will provide hands free and hand-held use and will help provide comfort in transportation. Multiple compartments must be another feature, both for specific purposes and for general use. Insulated sections to keep food and beverages cool, and waterproof lining for easy cleaning should be included too.

After listing the requirements, all of them were prioritized based on their relevance and then categorized as *must-have*, *should-have*, or *nice-to-have*. These requirements act as clear goals for the development and evaluation of the backpack concepts throughout the design process.

For a more detailed description of the Product requirement list, refer to the appendix chapter.

4.3. Develop

Throughout the “Develop” phase of the second diamond of the Double Diamond design process, a total of three iterations were conducted. A key part of this framework consists of creating feedback loops in the design process to test the ideas, evaluate them and refine them to achieve higher quality results. Consequently, some methods were repeated and applied in each iteration.

A - FIRST ITERATION

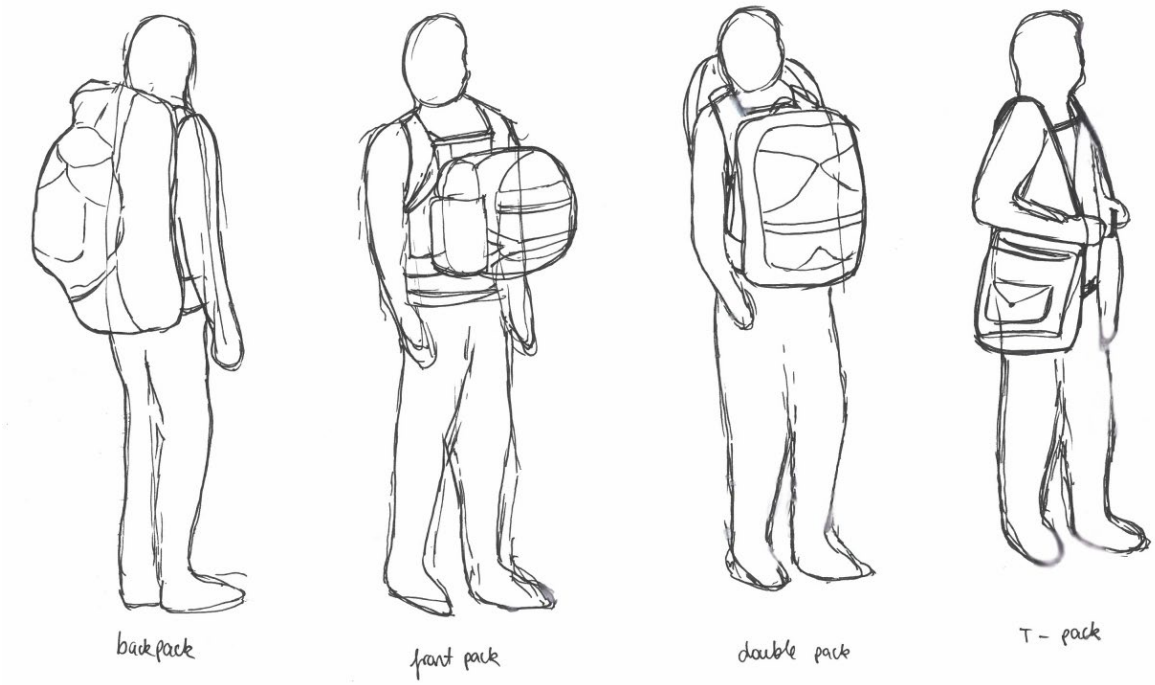
4.3.1. Brainstorming

The brainstorming method was used multiple times throughout the entire design process. Generating ideas using sketching as a tool helped develop innovative ideas and visualize the product’s evolution in a physical format.

Sketching was introduced already from the early design phases as a tool to support both research, and research through design. This process involved numerous iterations. It consisted of a diverging and converging process, where many ideas were sketched during brainstorming and then the selected ones were further developed.

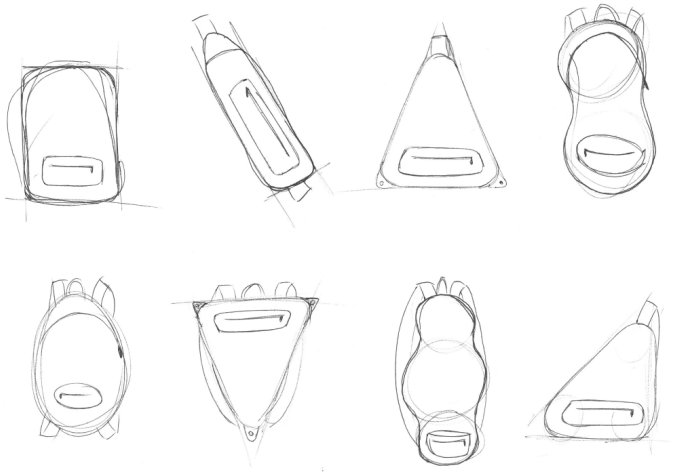
Exploring the different transport possibilities (Figure 9) through sketching was one of the first uses of brainstorming. The following iteration aimed at exploring different backpacks shapes, searching for the most suitable for the target user.

Figure 9: Types of bags



Based on the findings from the product competitor analysis and after the initial brainstorming, a traditional backpack was identified as the most appropriate solution for transporting items. While the general shape still had to be further developed, straight corners increased volume capacity, therefore, ninety-degree angles were considered most appropriated (Figure 10).

Figure 10: Backpack shapes



4.3.2. Try it yourself

Several bags with both functional and aesthetic features relevant to the intended product design were acquired (Figure 11). These backpacks included functions such as insulation for food and drinks, designated compartments or designs suited for harsh weather conditions. They were analysed and inspected to identify the most relevant aspects and details of their designs. Close observation of the material was also crucial.

Figure 11: Tested bags



After inspecting them from the inside and the outside, the bags were disassembled to analyse the internal components of the linings, as well as how the bags were produced. The type of stitches and the amount of fabric layers used were also studied in detail.

Zippers divide the different pockets effectively. For the compartments meant to carry food or drink, inside lining with waterproofed material is placed, which also allows easy cleaning. Foam inside the backpack walls provide stability and a sturdy look. The stitching is typically hidden within the interior layers. Velcro is a flexible way to provide modular walls.

The collected insights were used in the next steps of the concept generation process to create solutions that are desirable, viable and feasible.

Another way the “Try it yourself” method was implemented was by conducting a temperature experiment. It aimed to investigate how efficient the insulation compartments in lunch bags were.

The experiment involved measuring the temperature of different liquids before and after spending several hours at room temperature inside lunch bags (Figure 12). It compared how many degrees cooler the liquids remained in an insulated lunch bag versus a regular cotton backpack without insulation foam. Various liquid volumes and container types were tested.

The results showed that the temperature difference was only noticeable during the first four hours. After more than four hours both liquids reached approximately the same temperature. However, adding an icepack to the insulated compartment significantly increased the time the drinks stayed cold. Consequently, the greater the volume of cold liquid placed inside the compartment, the longer it stayed at a cool temperature.

For a more detailed description of the cooling experiment, refer to the appendix chapter.

Figure 12: Cooling experiment



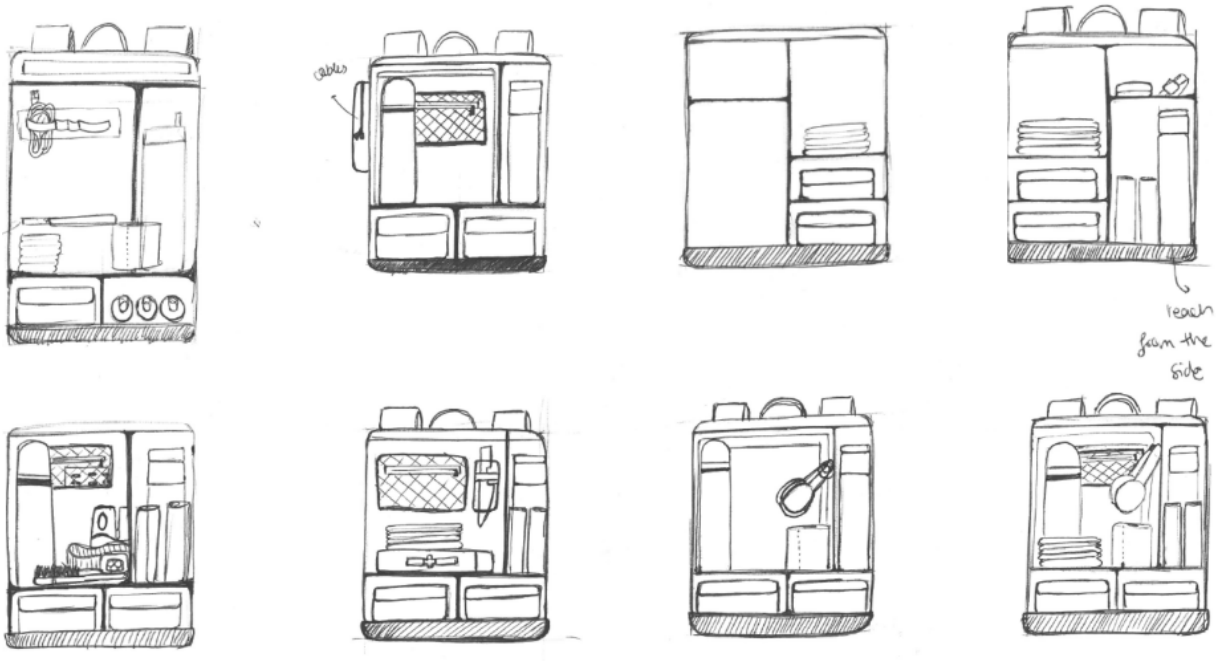
4.3.3. Concept generation

As previously mentioned, the idea generation process was iterative throughout the whole design, which led to multiple concept generation stages. However, to be able to develop a defined concept the arrangement of the compartments had to be established first.

The next iteration consisted in brainstorming about the storage possibilities, which compartments the backpack was going to have and which objects were going to be in them. Together with the conclusions of the requirements list of what needed to fit inside of the backpack, this iteration resulted in some concepts of different dispositions of the compartments.

Lunchboxes were the heaviest item in the bag, and therefore they had to be placed at the bottom of the bag, to achieve a balanced weigh distribution. Water bottles and cans also sum up heavy weights, and their placed influenced the weight distribution. Other smaller items like small tools, pills or batteries needed a storage system that would not make it difficult to reach. Coffee thermos, hearing protection or toilet paper were also brought by some of the users. In different compartments arrangements (Figure 13) are shown where usual items are placed in different pockets.

Figure 13: Compartments possibilities



In Figure 14 and Figure 15 various arrangements implemented into bag concepts are shown.

Figure 14: Backpack initial concepts 1

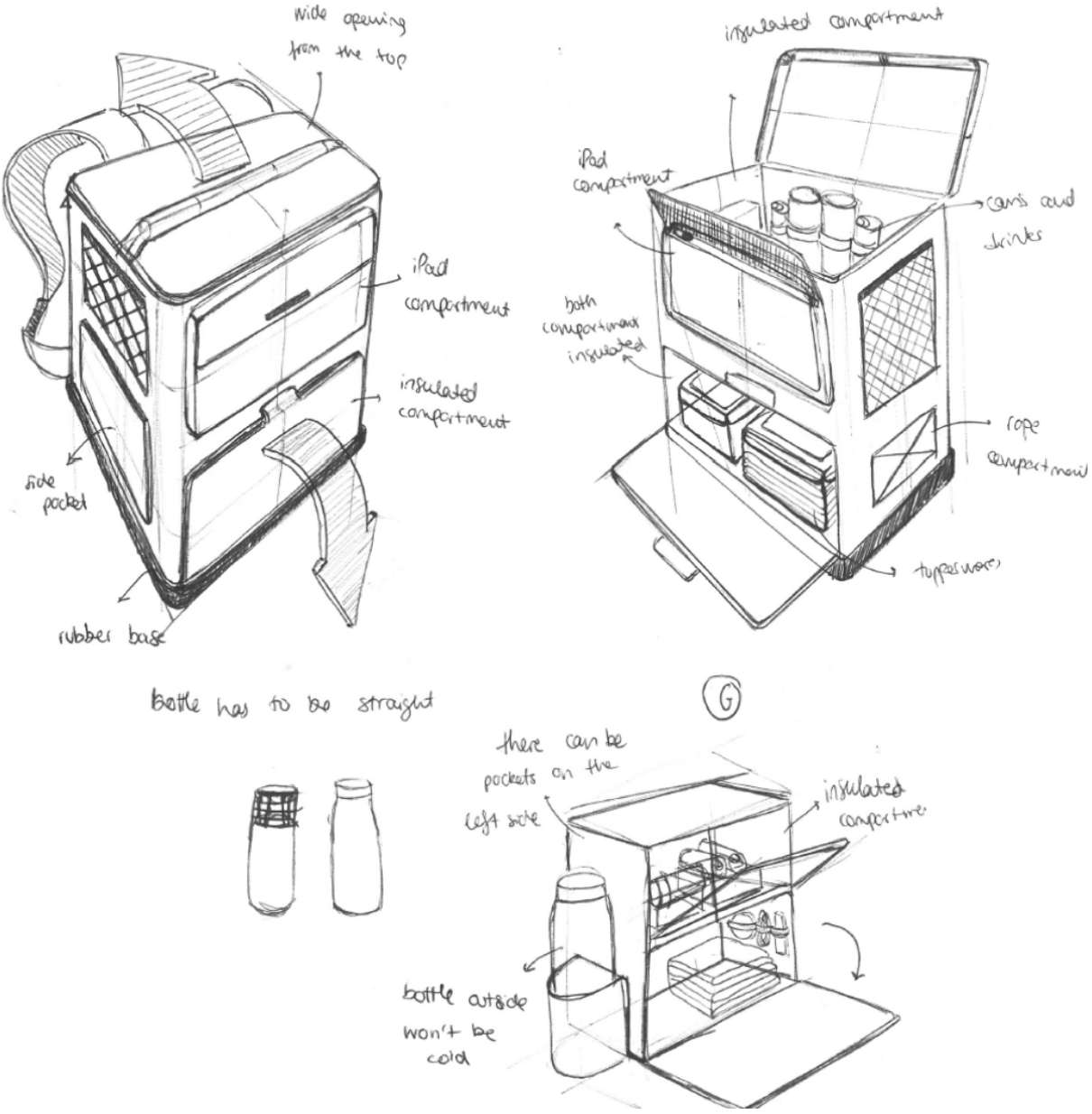
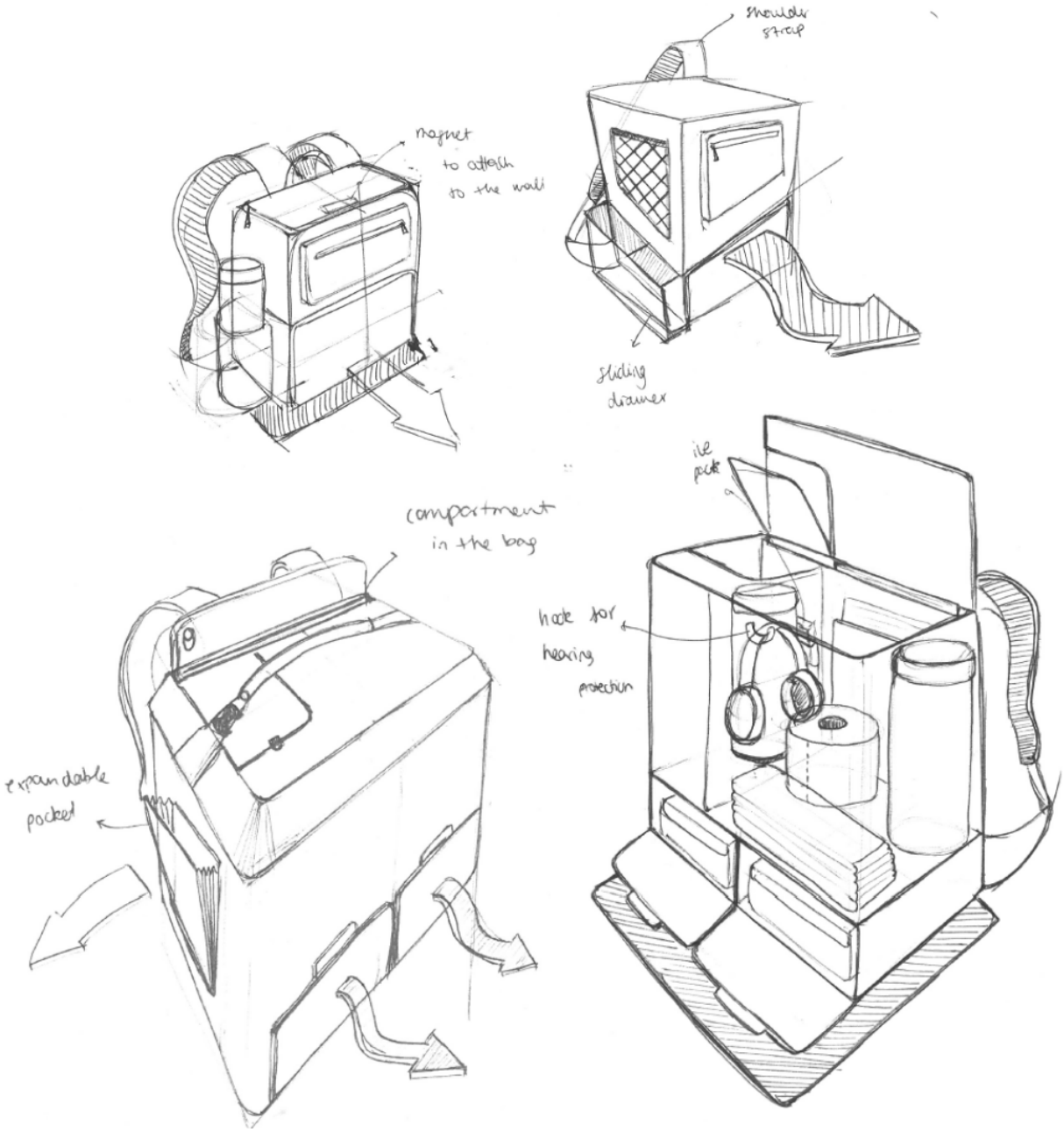


Figure 15: Backpack initial concepts 2



4.3.4. Mock-up

To test the different compartments arrangements, two of the most balanced concepts in terms of weight were selected. Two mock-ups with the two different dispositions were made to test out the most suitable one (Figure 16). Different aspects needed to be analysed: weight distribution, accessibility of compartments, and reachability from the compartments inside the cabin.

Figure 16: Backpack mock-up



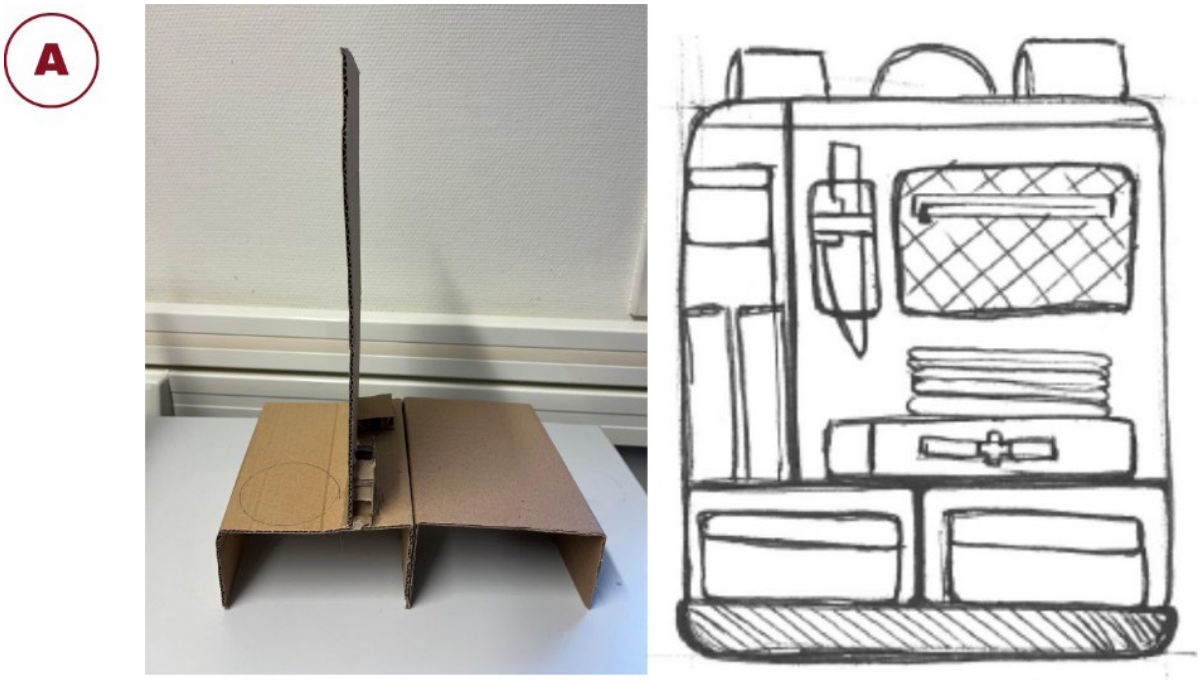
For testing the weight distribution, objects such up food lunchboxes, water bottles, clothes, tablets, and toilet paper were placed inside of the mock-up (Figure 17). To represent a real-world setting, some weights were placed inside the lunch box to simulate they are full of food and the water bottle was filled. The maximum weight would be 7 kilograms.

Figure 17: Mock-up with personal items



In option A (Figure 18) both lunchboxes are placed at the bottom of the bag and the water bottle, coffee thermos or cans are placed on the left side. The rest of the items are placed on the right side. This disposition relies on how heavy the usual extra objects are, since they have to compensate for the bottle weight. In this option, the water bottle can be reached from the top.

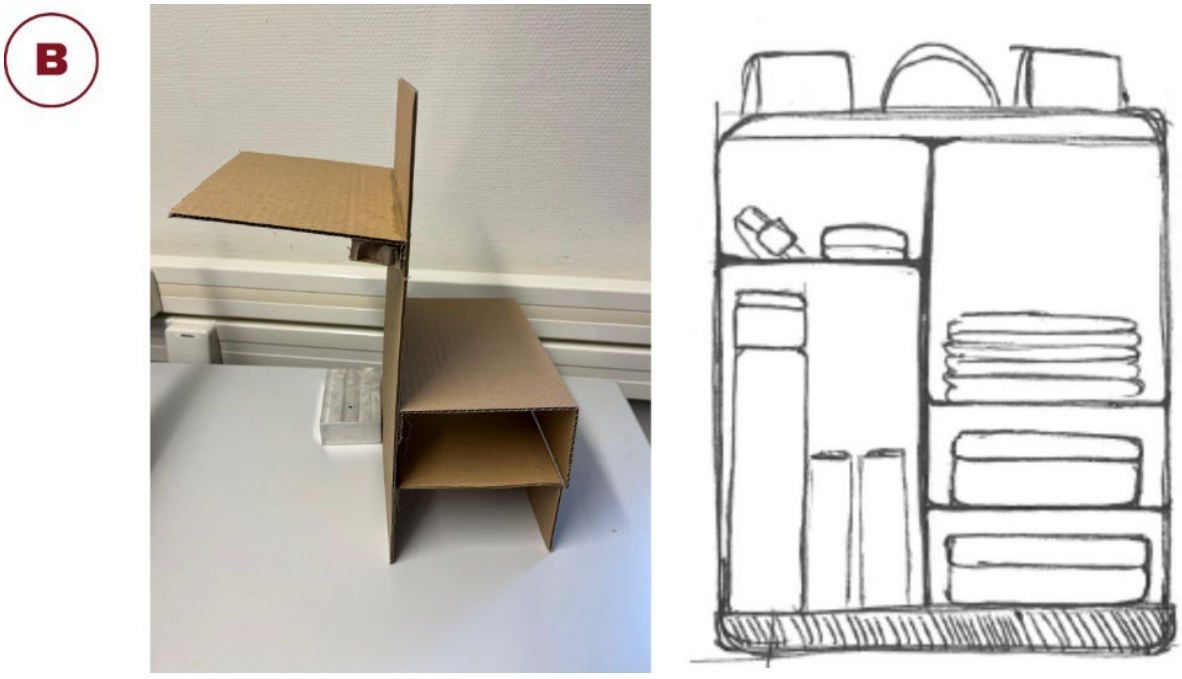
Figure 18: Mock-up disposition A



Option B (Figure 19) consisted of stacking both lunchboxes one on another and having the bottle on the left side. This way both items would balance each other's weight. In this option, the water bottle can be reached from the left side

In both arrangements, the tablet is placed in the back side, centered in the middle and close to the user's bag.

Figure 19: Mock-up disposition B



B - SECOND ITERATION

4.3.5. Concept user testing

The goal of the concept user testing was to find out which of the proposed compartments arrangement was more suitable for users, in terms of reachability and ergonomics. The test consisted of providing a cardboard mock-up to users in a real setting, TERRI's machine cab, and asking questions about comfortability (Figure 20).

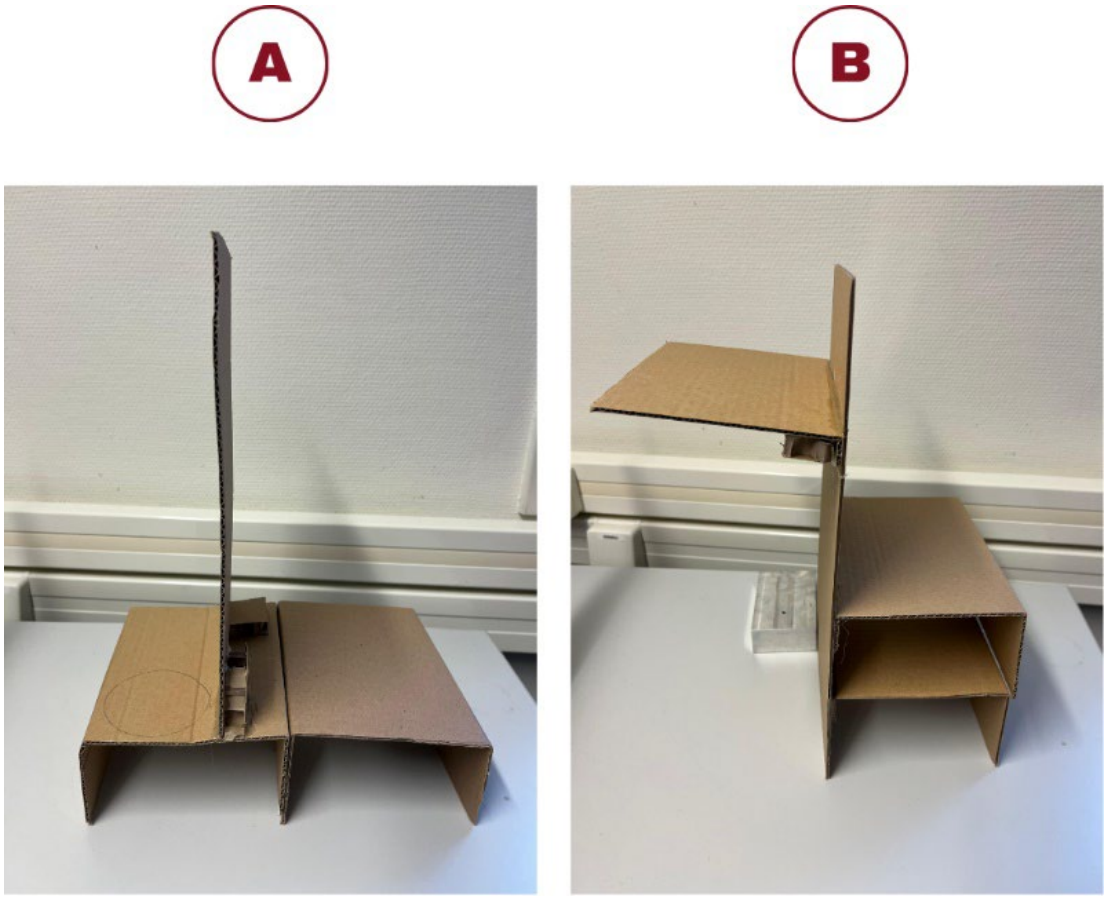
Figure 20: User testing mock-up in the cabin



The setting consisted of a real Terri cabin that was unequipped. However, using Isac Ragnarsson thesis "Working comfortably in the woods" mock up helped have a real world setting of the future Terri's Forest machine interior design. According to Ragnarsson (2025) the interior had a designated spot for the backpack at the front right of the machine. That place had a ninety-degree corner with enough space to fit the bag. The wall was ferromagnetic which will allow the magnet in the lid to get attached to the wall when the user wants it open. Under the tray that stays on top there will be LED lights that will illuminate the bag to allow easy sight inside of the bag.

The mock-up had two versions of arrangements, A, and B (Figure 21). The test consisted of asking 4 different potential users with different body measurements about the accessibility of the mock-up pockets.

Figure 21: Mock-ups arrangements



The user test indicated that in the first option, B, users found it easy to access the lunchboxes compartments, since the access was higher, in comparison to A which they found more difficult to reach. The water bottle was faster and easier to reach from the side (option B) rather than from the top (option A). Lastly, the weight distribution resulted in being more balanced in option B, since water bottle and lunchboxes were compensating each other.

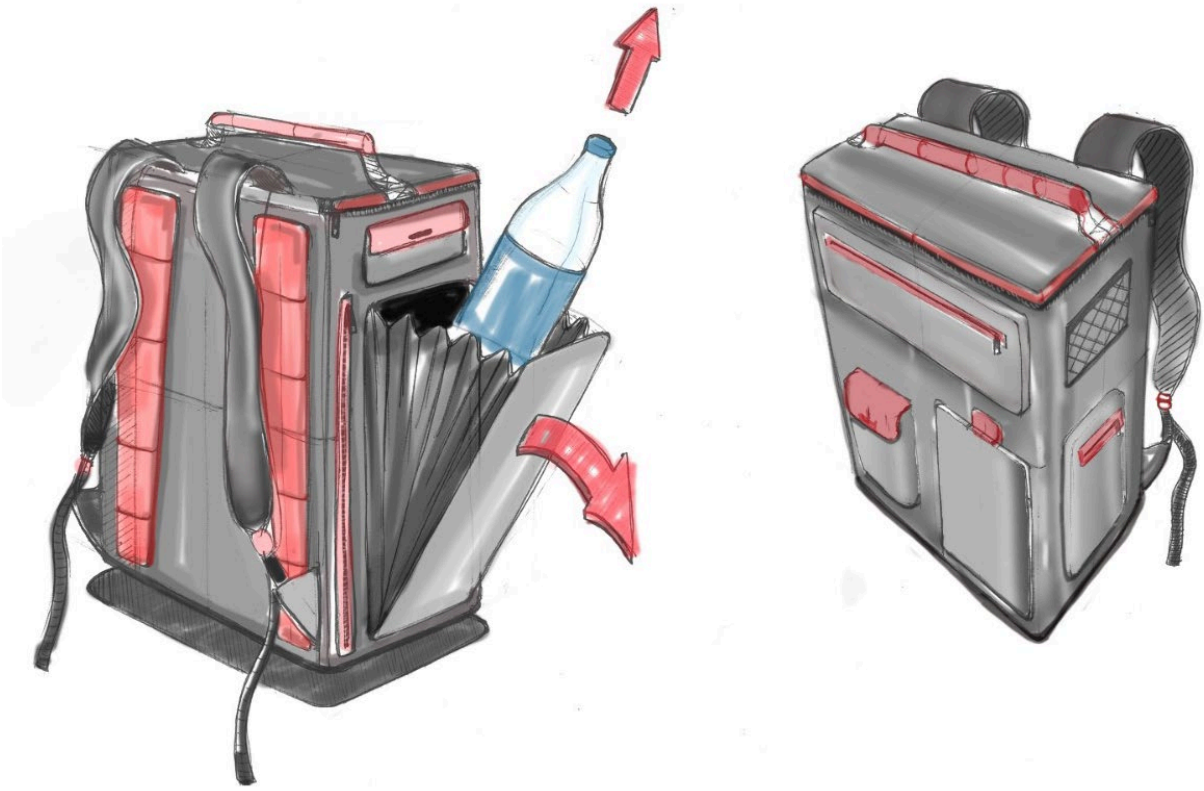
The test resulted in the B version being the winner for every user that tried it. Some relevant suggestions were also provided by the users, such as the water bottle compartment could be easier to access if it opened with an angle.

To explore further information about the user test, refer to the appendix.

4.3.6. Concept generation

After making a decision of which compartments distribution was the selected one, a rendered sketch was made (Figure 22). In this sketch the storage is placed following option B, where the lunchboxes are sacked on the bottom right and the bottles can be reached from the side. This pocket only allows opening until a certain angle, to prevent spilling the bottle.

Figure 22: Backpack concept



4.3.7. Mock-up

Once a choice of the main shape, aesthetic and storage was made, a second mock up with textile was created. In this case, actual fabrics were used, and zipper and textile hardware were also implemented, such as Velcro, rubber bands or mesh fabric. This helped to get better understanding of how these materials behave and what feelings and experience they transmit (Figure 23).

Figure 23: Textile mock-up



Creating this mock-up helped get knowledge about sewing, stiches and how fabrics behave (Figure 24).

Figure 24: Fabric tests



The mock-up was placed inside the cabin again to check the measurements and the accessibility to the different compartments (Figure 25).

Figure 25: Mock-up placement

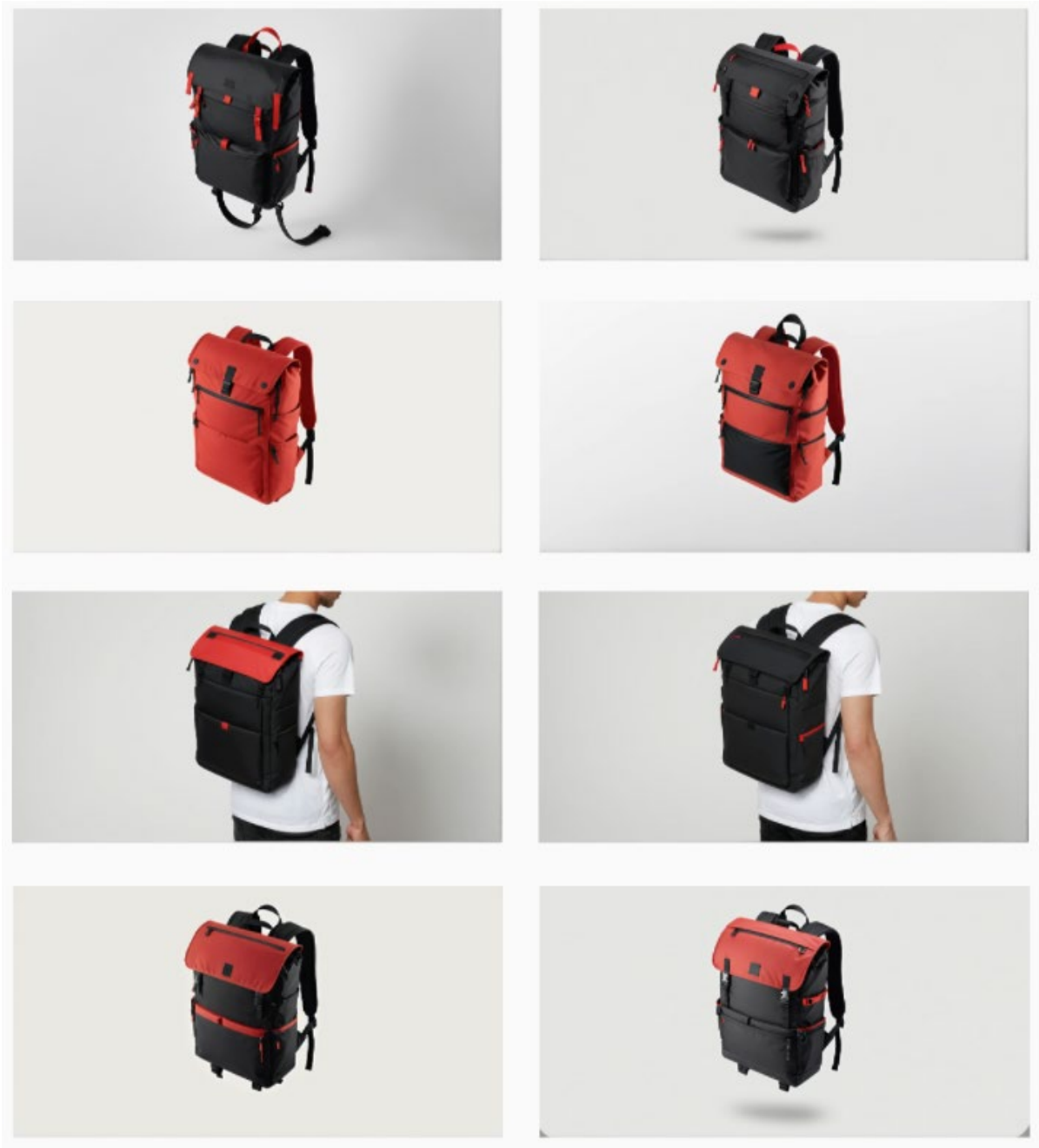


C - THIRD ITERATION

4.3.8. Concept generation

Having decided on the main shape and look and the storage system, it was time to refine the aesthetics details. To get help with a generation a large number of varieties, the AI tool Vizcom was utilized. Starting from the initial shape and description of Figure 22: Backpack concept, numerous possible concepts were developed. The focus was mainly placed on the aesthetics and material finishes. It was important to look at the connection between fabrics, zippers and closing mechanisms and pocket details (Figure 26).

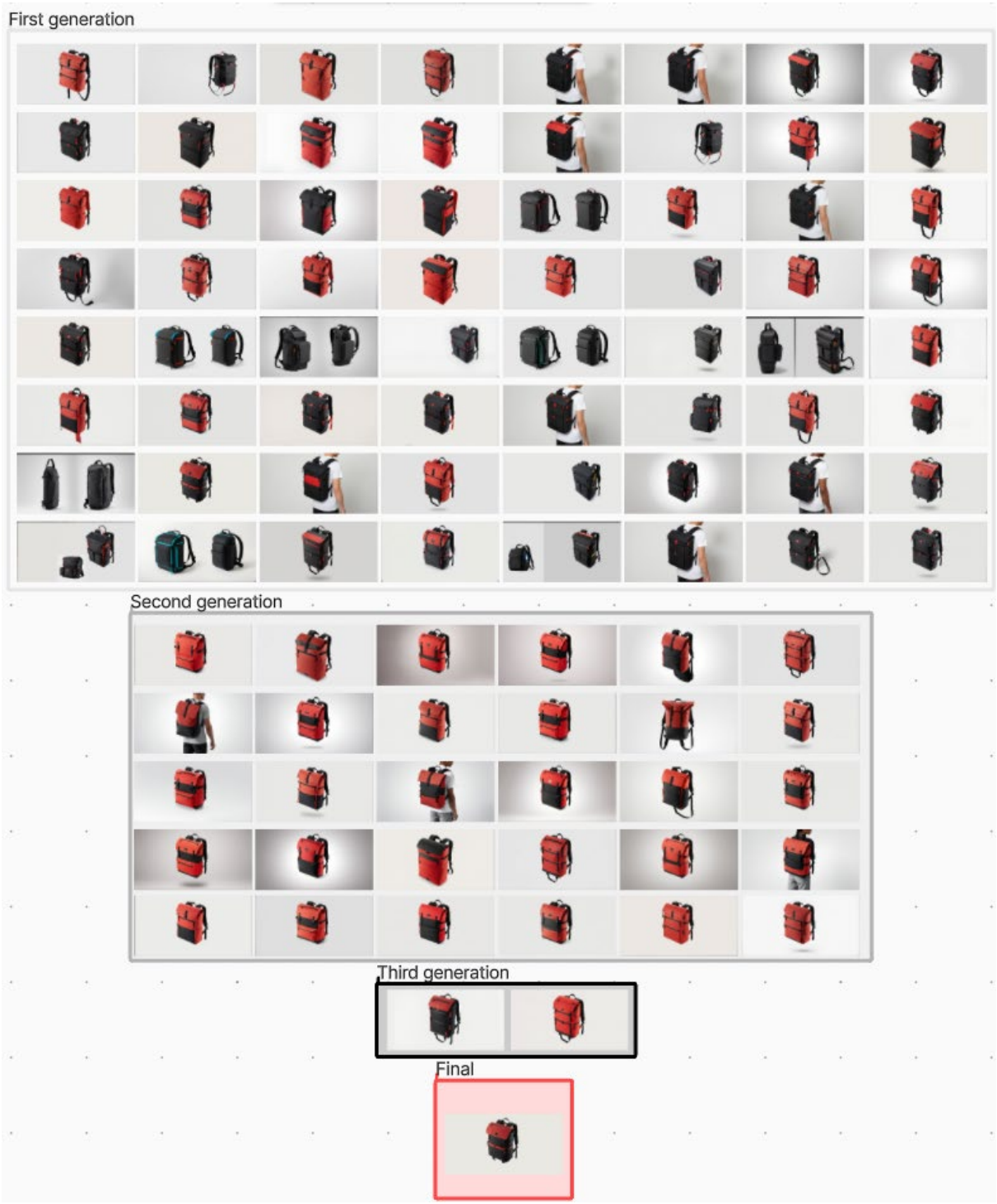
Figure 26: Backpack concepts



4.3.9. Concept screening

Because of the high generation of different ideas, screening to find the most suitable concept was needed. This process was again iterative, since from the selected variations, further concepts were developed. Some of the winning ideas were grouped for the next phase and further concepts were defined until the final and ideal concept was chosen (Figure 27).

Figure 27: Concept screening



Some variations of the final concept were selected and further defined in photoshop, changing small details, colours and adding the Terri logo (Figure 28).

Both options were shown to the company and the second one (Figure 29) was the one selected for several reasons. The bag will be exposed to harsh conditions and mud which would make it likely to get dirty easily. Because of that, having the colour black as the main colour is smarter so that it will not look dirty as easily.

Figure 28: Concept 1 & 2



Figure 29: Concept 3 & 4



4.4. Deliver

4.4.1. Prototyping

The final concept was developed into a prototype (Figure 30) after several iterations. The main goal of the final prototype was to demonstrate the materials and finishes of the product using the actual selected fabrics. The function of the different compartments and pockets was also a main reason to show the design in a physical prototype. Creating a backpack from real textile materials made it easier to appreciate the texture and tactile feeling. It also reflected the emotional response users might have when interacting with the product.

Figure 30: Prototype front



Some of the materials used were cotton canvas, polyethylene foam (PE Foam), recycled polyester fabric and waterproof coated textiles. Hardware components such as zippers and buckles were implemented to achieve deeper details.

However, before developing the final prototype, several tests with the chosen materials were conducted. It helped determine the most effective ways to achieve the final product. Various sewing technics needed to be explored to achieve a high-quality result. Firstly, the patterns of the fabric needed to be defined with the correct measurements and guides. A sewing machine was the primary tool used in assembling the prototype (Figure 31).

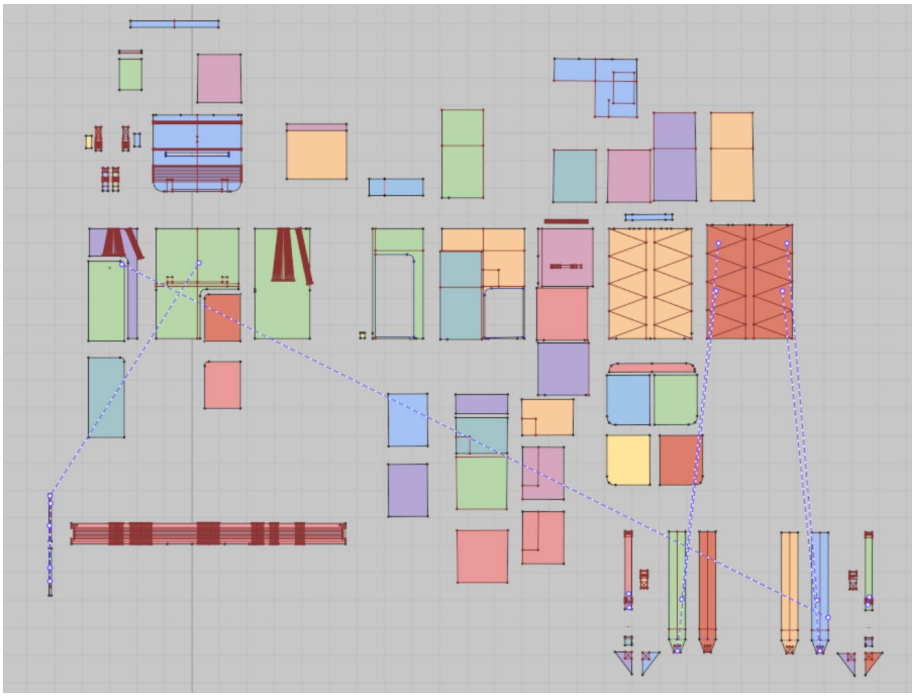
Figure 31: Prototype parts



The final version is a functional prototype that also reflects the product's aesthetic and final look, except for some minor colour details.

Additionally, another tool was used for developing the prototype, 3D modelling software CLO 3D was used to create a CAD version of the product. This digital visualization of the product was helpful for generating accurate patterns of the pieces needed for real-life construction (Figure 32).

Figure 32: Prototype patterns



5. Final Product Results

5.1. Product

The final product is a multifunctional bag designed to fulfil forestry machine operators' needs. It serves operators both during working time and off duty hours. The bag can be integrated into the cabin and serve as one more storage possibility for the users. Furthermore, it functions as a merchandising product for Terri, helping promote the company's image (Figure 33).

Figure 33: Contextual render 1



The primary design focus of the backpack was on compartmentalization, linked to the operators' frustration of having chaotic bags where items were difficult to find. The bag consists of four main interior compartments and two additional exterior pockets. The main interior compartments' purpose is for storing food and beverages. In the bottom right corner, there is an insulated compartment lined with a waterproof fabric, specifically for storing lunchboxes. In case of any spill or leak, the bag can be easily cleaned thanks to the inside lining. The pocket opens with a zipper (Figure 34). The whole bag is covered by a big lid (Figure 35).

Figure 34: Lunchboxes compartment



Figure 35: Contextual render 2



On the left side, there is a compartment destinated to carrying drinks (Figure 36). Water bottles or cans can be placed in that pocket, which is also insulated to keep a cool temperature. Additionally, there is space to include an ice pack for users who prefer colder drinks. Both pockets are accessible with a zipper, which also prevents items from falling out (Figure 37).

Figure 36: Drinks side compartment



Figure 37: Water bottle compartment



The two insulated compartments are easily reachable from the front when the bag is placed inside the cabin (Figure 38). The goal of this layout is that the operator can conveniently access their drinks and return in the bag to keep it cool. The lunchboxes can also remain inside the bag until lunch time, to maintain their temperature until needed. On the front and the bag of the bag there are two pockets in which extra items can be stored (Figure 39).

Figure 38: Insulated compartments



Figure 39: Front compartments



Another item that operators frequently carry during working hours was electronic devices like tablets, according to the questionnaire conducted. During the mock-up test, the tablet was one of the heaviest items. As a result, it was decided to place the tablet compartment on the back side of the backpack, closer to user's back (Figure 40). This placement helps ensure there is not any potential damage and promotes balanced weight distribution. The pocket is accessible from the left side when the bag is placed inside of the cabin and has padded walls for supplementary protection.

Figure 40: Tablet compartment



In addition to the specialized compartments, the backpack also offers extra compartments that are meant for general use (Figure 41). They do not have a defined purpose, allowing users to adapt them based on their specific needs. Items such as spare clothes, tools, toilet paper, or smaller essentials like batteries or pills can also be stored in these areas.

Figure 41: General use compartments



A removable interior divider is attached with Velcro (Figure 42) and helps create different modular sections that can be arranged according to user preferences. This flexibility increases functionality and offers numerous possibilities for the user to personalize the interior to their wants.

Figure 42: Velcro divider



In Figure 43 the back side of the backpack is shown. It has padded foam to provide a comfortable fit on the user's back when carried to and from the cabin. It also features two shoulder straps, enabling hands-free transport.

Figure 43: Back padding



The bottom of the bag is made out from a more rugged material and includes four base points that act as feet, preventing the bag from getting dirty when placed on the ground (Figure 44).

Figure 44: Bottom supports



The red lid that covers (Figure 45) the entire top part of the bag is made from a waterproof material, recycled polyester, and serves two purposes. First, it protects the bag from the rain due to its water- repellent properties. Additionally, it functions as a merchandising product, since it shows the brands colours along with the logo. Since the bag will be used in forestry machine environments, it may also be seen by other machine operators, helping promote brand visibility.

Figure 45: Red lid view



The colours chosen for the backpack are black and red, which matches the shade of red used on Terri's machines. Initially, both colours were intended to be used equally. However, it was later decided that lack should be the dominant colour as it is more visually appealing to user and less likely to show dirt than red.

To help users identify the different compartments, the zippers, that are mostly black, feature a red detail on the puller (Figure 46). This visual sign not only relates to Terri's colour but also works as an example of semiotic design thinking, guiding users to find the pockets intuitively. Some final details were added to increase the product quality. Double stiches with red thread were featured next to the zippers.

Figure 46: Zipper pullers



One of the main challenges was the placement of the bag inside the cabin. In this case, the designated space in Terri's cabin is on the front right (Figure 47). It is placed under a light source, so it is easier to find and reach the items inside the backpack (Figure 48). The bag's lid stays open when the operator wants to access the inside, since it has a magnet that attaches it to the metal cabin wall. There is enough space for the operator to rotate the chair freely without stepping on the bag, since it is strategic placed on the corner. The bag will not move from that spot thanks to the bottom supports it has, a rubber matt that there is on the floor of the cabin and the magnet (Figure 49).

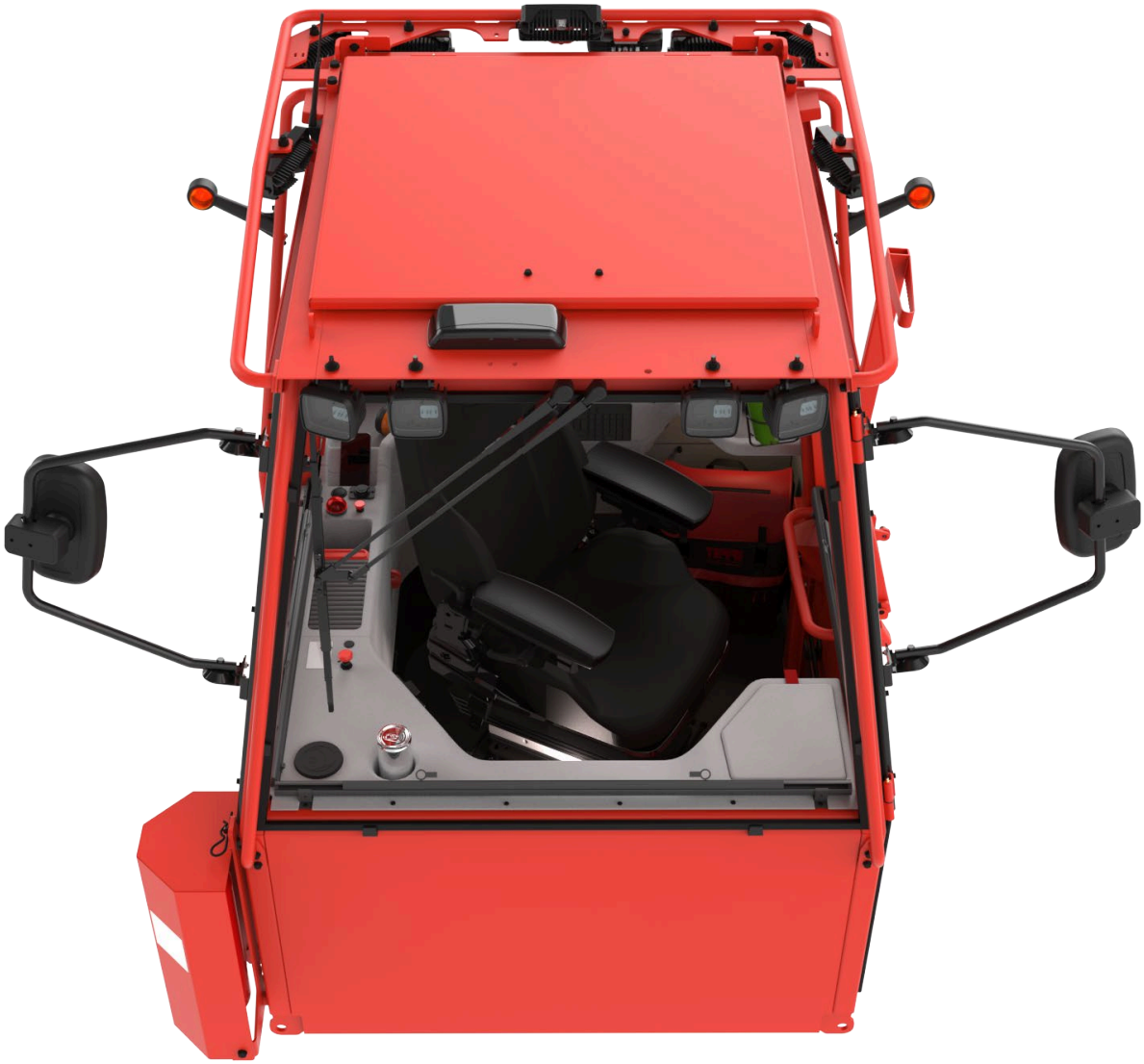
Figure 47: In-cabin bag placement view 1



Figure 48: Illuminated bag placement



Figure 49: In-cabin bag placement view 2



This bag is the result of the problems defined at the beginning of the project and improves forestry machine operators working experience (Figure 50).

Figure 50: Contextual render 3



6. Discussion and Conclusion

The aim of the project was to design a bag that could solve the problem of forestry professionals' who need to transport everyday objects from their homes to their cabins. Additionally, this bag also needed to be integrated into the design of Terri's cabins.

6.1. Discussion of research questions

- WHAT ARE THE EXISTING CHALLENGES FOR FORESTRY MACHINE OPERATORS WHEN CARRYING THEIR BELONGINGS?

From the several research methods conducted, various conclusions were drawn. It was observed that, in most of the cases, operators bring similar objects to their work shifts, food drinks, electronic devices, along with other objects that vary depending on the individual.

Their biggest challenge was having all items stored in one large compartment in their current bags, making organization of it very difficult and led to clutter. They expressed a need for compartmentalization as it would make the bags more organized and practical.

The second most significant issue found was not having cool drinks available during working hours, which can be particularly unpleasant during the hottest months. Drinks, or sometimes food, leaking was another related problem that was occurring when transporting bottles or containers in the bag. Insulated compartments with waterproofed inner linings helped solve these problems effectively.

Additionally, issues were also observed not only when carrying their bags but also when using them inside the cabin. In many cases, the bag was also used as an additional form of storage. However, having a cramped interior and difficulties in accessing certain items during work hours were considered major inconveniences by machine operators.

Another common issue was that their bags were not stiff enough to stay in place when placed inside the cabin. As a result, they could fall over and make it difficult to access the contents. The lack of a dedicated space for the bag inside the cabin was a challenge for the users, as often it had to be placed wherever there was room, sometimes interfering with the operator's work.

- WHAT MATERIALS ARE BETTER SUITED FOR A BACKPACK DESIGN THAT PRIORITIZES SUSTAINABILITY, MEETS USERS' NEEDS, AND WITHSTANDS THE CONDITIONS FACED BY FORESTRY MACHINE DRIVERS?

Based on the material analysis conducted during this project (section 4.1.2), recycled polyester (rPET) was identified as the most suitable material for backpacks designs for forestry machine operators. This decision was made considering five critical aspects: durability, weight, weather resistance, cost and environmental impact.

This choice of materials was based on three main considerations:

Sustainability: Sustainability was an important factor in the material selection. rPET was chosen for having a lower environmental impact compared to other materials options. Thermoplastic polyurethane (TPU) was selected for waterproofed fabrics due to its long lifespan and durability.

User needs: Addressing user needs was another critical point to consider when selecting the better suited materials. Users emphasized the importance of water resistance, ease of cleaning, lightweight materials and long-term durability. For this reason, a combination of rPET with TPU waterproof coatings was selected. Additionally, waterproof inner lining was also selected to facilitate the inside cleaning in the case if spills or leaks.

Harsh forestry conditions: The materials had to withstand demanding conditions. This includes the use of waterproof zippers that would perform best in the conditions addressed. And stiff structural elements made from PE foam. Although PE is still a fuel-based material, it can be sourced from industrial waste or recycled packaging, which helps reduce its environmental footprint.

One important factor in material selection is transparency of manufacturing and materials. This product promotes choosing production methods that are done locally in Sweden, avoiding long distance shipment of materials that can have a high environmental impact. Being transparent about material choices, as well as considering ethical values during the manufacturing, will lower the environmental impact.

- HOW TO DESIGN A BAG THAT SUPPORTS EFFECTIVELY FORESTRY MACHINE OPERATORS DURING BOTH WORK HOURS AND OFF DUTY TIME?

This research question reflects the design process behind the final product. The design development was based on interpreted findings from the research phase. User needs were considered so that the bag could fulfil the operators wants. They expressed the need of a bag that could store both work-related and personal items. It had to be comfortable to carry and be able to withstand the environment conditions, while also fitting within limited space inside of the cab. Lastly, it needed to align with the aesthetic preferences of the focus group.

To effectively support machine operators, a bag with related features was developed. Several aspects were implemented, including modularity, compartmentalization and versatility. The design features included insulated food pockets, durable and waterproof materials and neutral aesthetics. The bag offered functionality to withstand harsh work conditions and adaptability to be easily carried and placed inside the cabin.

Integrating elements such as detachable dividers for customizable sections, padded pockets for electronics, and insulated areas for food helped align the bag with users' preferences. Overall, appearance remains neutral offering operators to carry the bag confidently also during off-duty hours.

The development of a functional backpack demonstrates how the different functions were integrated, as well as reflects the overall aesthetic and user experience when interacting with the bag.

6.2. Methods

During the previous research conducted by the Enterprise course students, it was observed that a bag was a common item that was brought every day by the operators and proved to be very useful for them. While the decision to design a bag had already been made beforehand, the specific type of bag that was going to be developed was not determined yet. Starting the design process in the early stages of the project was beneficial for exploring different ways of transporting items. After the initial research, it was decided that a backpack was the most suitable option, and the project proceeded in that direction.

The decision to use the double diamond design methodology was beneficial because of being an iterative process. The iterations during the develop phase of the double diamond were helpful in refining the concept to a deeper level, as continuous adjustments were made. Three iterations of the double diamond were completed to reach the final result, which was time-consuming, but necessary for the project.

Besides the usage of double diamond, a human-centered approach and design thinking principles were used. Including users in the process and letting them contribute was beneficial to developing a product that could solve issues people were currently facing.

The questionnaire was distributed through to Swedish Facebook groups in which the focus group was involved. This approach was very effective, as users showed strong interest in the project and felt it was a problem they wanted to see solved. The questionnaire got significant attention, and 135 answers were collected in less than a week. Because of the high user engagement, some of the users even contacted the author directly which led to having closer contact with the operators. These conversations, along with shared images and detail descriptions of their current way of transporting items were gathered and provided helpful insights for the later development stages.

For validation of the final delivery of the product, the backpack was revised to check if it was fulfilling the user needs and the requirements that were defined during the earlier stages. Comparing the requirements to the insights collected during the questionnaire and conversation with the users validated that the user needs were correctly fulfilled.

The user test was another way of validation of the design, to see if the features included in the bag were actually fulfilling users' preferences.

Documenting every method's approach and its implementation in the report helps ensure the project's reliability, as it allows to be understood in the future by observing the steps.

6.3. Conclusion

This thesis aim was to explore how a multifunctional bag could be designed to support forestry machine operators both during work and in off-duty situations. The objective was to find a balance between usability, durability and visual versatility when developing a concept solution that considers the conditions of the specific forestry work context.

By conducting user research, competitor analysis of existing products and iterative development stages, essential features that needed to be included were identified, such as flexibility, compartmental organization, comfort, waterproof materials, and modular components.

Furthermore, the product creates a connection between the user and the bag, since its interaction can also be connected to emotions. The functional bag design becomes also impactful, memorable, and meaningful design that improves the operators working experience.

The final result reflects a design concept that integrates professional necessities with personal usability, showing that it is possible to meet the tough conditions of forestry environments without compromising portability or visual appeal. This project offers value to the field of human-centered design for specialized professional users.

6.4. Personal reflection

The creation of different types of mock ups and prototypes both with easy materials like cardboard and fabric helped get physical understanding of the proportions and feeling the materials transmitted (Figure 30). However, due to the lack of previous knowledge or experience of the author working with sewing machines and fabric work, the process was challenging and required numerous preliminary tests, trial-and-error learning and online tutorials. The final prototype did not have the expected final details due to limitations with the type of fabrics and resources. The fabrics used and the colours are not exactly the expected ones. The Terri logo was thought to be placed in the bag by embroidery, but instead it was a glued patch. Additionally, the usage of a new software unfamiliar to the author took extra time, which was initially not planned in the schedule. Despite these challenges, the results were of high-quality, and the details were successfully captured in both the final prototype and the rendered visuals.

I would like to thank my classmates in the studio for being there throughout this thesis journey, and to Moe and Ulrik for their help and advice.

Figure 51: Cabin with cardboard mock-up



6.5. Further research

To position the current product into a comparable market category and develop a proper business model, cost comparison and pricing must be taken into consideration in the future. In this project, the prototype was elaborated using handcrafted methods, without any industrial production processes. The creation of the bag was done by a single individual, which allowed attention to small details but is not feasible for mass production. The cost of a handmade bag could be very high, which would not be competitive in the market. Further research should include the consideration of the production cost, actual manufacturing methods and the price of wholesale materials.

For further development, the final prototype could be tested under real working conditions, such as inside a forestry machine cabin. Field testing with an operator over a full workday would be helpful to observe the pain points it has and improve them for a final iteration. However, this project's scope did not cover this and there was no time for a last iteration. It would also provide valuable insights to evaluate the backpack being used in different forest machines and beyond Terri's cab, as well as testing it in everyday settings, not work related, to test its flexibility. Moreover, the project lacked long-term field testing, which could support further design development.

As an additional form of design validation, the emotional response the bag evokes in the user and the relationship it is created between product and user could be tested. Exploring the emotional connection generated through the user's interaction with the bag could help prove if the bag fosters a meaningful engagement. To evaluate this, an empathy map could be created to capture the relationship and the emotions between the bag and the user. This map captures the feelings, thoughts and actions of the user can be collected and analysed to later evaluate whether a genuine emotional connection has been formed.

7. References

Bonnardel, N., & Didier, J. (2020). Brainstorming variants to favor creative design. *Applied Ergonomics*, *83*, 102987. https://doi.org/https://doi.org/10.1016/j.apergo.2019.102987

British Design Council. (2025). *The Double Diamond Design Process*. Retrieved from https://www.designcouncil.org.uk/our-resources/the-double-diamond/history-of-the-double-diamond/

Castro, G. a. M. J. a. N. L. a. M. R. (2016). *Machinery and Equipment in Harvesting*. https://doi.org/10.1007/978-3-642-54601-3_183

Căşeriu, B., & Blaga, P. (2023). Considerations Regarding the Ergonomics of the Vehicle Interior. *Acta Marisiensis Seria Technologica*, *Vol. 20 (XXXIV)*, 2023. https://doi.org/10.2478/amset-2023-0015

Dam & Teo, R. F., Yu Siang. (2024). *What is Design Thinking and Why Is It So Popular?* Retrieved 2025, January 27 from https://www.interaction-design.org/literature/article/what-is-design-thinking-and-why-is-it-so-popular

Dvořák, J., Kováč, J., & Krilek, J. (2020). *Ergonomic Operational Working Aspects of Forest Machines*. Cambridge Scholars Publishing.

Elliott, K. C., Lincoln, J. M., Flynn, M. A., Levin, J. L., Smidt, M., Dzugan, J., & Ramos, A. K. (2022). Working hours, sleep, and fatigue in the agriculture, forestry, and fishing sector: A scoping review. *Am J Ind Med*, *65*(11), 898-912. https://doi.org/10.1002/ajim.23418

Fazeli, H. R., & Peng, Q. (2022). Generation and evaluation of product concepts by integrating extended axiomatic design, quality function deployment and design structure matrix. *Advanced Engineering Informatics*, *54*, 101716. https://doi.org/https://doi.org/10.1016/j.aei.2022.101716

Gibbons, S. (2018). *Empathy Mapping: The First Step in Design Thinking*. Retrieved 2025, April 05 from https://www.nngroup.com/articles/empathy-mapping/

Heder Brandt, P., Olsson, A., Dahlquist, K., & Inal, T. (2023). "Profitability is sustainability:" framing of forest management practices by the Swedish forest industry. *Scandinavian Journal of Forest Research*, *38*(7-8), 429-441. https://doi.org/10.1080/02827581.2023.2252740

Hekkert, P., & Karana, E. (2014). Materials Experience. In R. Elvin Karana and Owain Pedgley and Valentina (Ed.), *Materials Experience* (pp. 3-13). Butterworth-Heinemann. https://doi.org/https://doi.org/10.1016/B978-0-08-099359-1.00001-1

Henriksson, F. (2021). *On Material selection and its consequences in product development*. Linkopings Universitet (Sweden).

Ilstedt, S. (2002). Semiotics in product design.

Karana, E., Hekkert, P., & Kandachar, P. (2008). Material considerations in product design: A survey on crucial material aspects used by product designers. *Materials & Design*, *29*(6), 1081-1089. https://doi.org/https://doi.org/10.1016/j.matdes.2007.06.002

Karwowski, W., Soares, M. M., & Stanton, N. A. (2011). *Human factors and ergonomics in consumer product design: Uses and Applications*. CRC Press.

Keenan, R. a. K. J. P. (1993). The ecological effects of clear-cutting. *Environ.Rev.*, *1*, 121-144. https://doi.org/10.1139/a93-010

Krippendorf, K. (1989). Product Semantics. A Triangulation and Four Design Theories. 2nd Conference on Product Semantics, University of Industrial Arts Helsinki.

Krippendorff, K. (2006). *The semantic turn. A new foundation for design*.

Lammers, J. (2021). *Empathy Mapping: Bridging Cultural and Linguistic Divides in International Online Education*. https://scholar.uwindsor.ca/itos21/session3/session3/4/

Lockwood, T. (2009). *DesignThinking. Integrating Innovation, Customer Experience, and Brand Value*.

Lugt, R. v. d. (2005). How sketching can affect the idea generation process in design group meetings. *Design Studies*, *26*(2), 101-122. https://doi.org/https://doi.org/10.1016/j.destud.2004.08.003

Matteo, A. (2023). *Evolving Trends in the Cabin Design for the Industrial Heavy Machinery*. Retrieved 2025, January 23 from https://www.heavyquipmag.com/2023/12/22/evolving-trends-in-cabin-design-for-the-industrial-heavy-machinery/

Michael, P. (2009). Chapter 2 - Design Requirements. In P. Michael (Ed.), *Materials Enabled Designs* (pp. 23-50). Butterworth-Heinemann. https://doi.org/https://doi.org/10.1016/B978-0-7506-8287-9.00002-1

Milton, A., & Rodgers, P. (2023). *Research methods for product design*. Hachette UK.

Monö, R. (1997). *Design for Product Understanding*. Liber.

Moreau, G., Chagnon, C., Achim, A., Caspersen, J., D'Orangeville, L., Sánchez-Pinillos, M., & Thiffault, N. (2022). Opportunities and limitations of thinning to increase resistance and resilience of trees and forests to global change. *Forestry: An International Journal of Forest Research*, *95*(5), 595-615. https://doi.org/10.1093/forestry/cpac010

Norman, D. A. (2013). *The Design of Everyday Things: Revised and Expanded Edition*. Basic Books.

O'Donoghue, J. (2023). *Desirability, Viability, Feasibility Framework: A Practical Guide*. Retrieved April 19, 2025 from https://makeiterate.com/ideos-desirability-viability-feasibility-framework-a-practical-guide/

Parajuli, M., Hiesl, P., Smidt, M., & Mitchell, D. (2020). Factors influencing productivity and cost in the whole-tree harvesting system. *Clemson Cooperative Extension, Land-Grant Press by Clemson Extension*, *1079*, 1-8.

Ragnarsson, I. (2025). Working Comfortably in the Woods. Redesigning the interior of a forestry machine cab to enhance aesthetic, storage capacity and user experience.

Santharm, B. A., & Ramanathan, U. (2022). *Supply chain transparency for sustainability – an intervention-based research approach*. https://www.emerald.com/insight/content/doi/10.1108/ijopm-11-2021-0684/full/html

Sisay, A. W. a. H. Y. (2016). Multi-position ergonomic computer workstation design to increase comfort of computer work. *International Journal of Industrial Ergonomics*, *53*, 1-9. https://doi.org/https://doi.org/10.1016/j.ergon.2015.10.005

Soares, M. M., & Rebelo, F. (2017). *Ergonomics in Design. Methods & Techniques*.

Sun, Z., Doiro, M., Sá, J. C., & Santos, G. (2023). Shaping the Conscious Behaviors of Product Designers in the Early Stages of Projects: Promoting Correct Material Selection and Green Self-Identity through a New Conceptual Model. https://doi.org/10.3390/su151914463

Sunde, H. (2018). *Product Semantics. How design can affect user response and behaviour* Norwegian University of Science and Technology. https://www.ntnu.edu/documents/139799/1279149990/28%2BArtikkel%2BHenrik%2BSunde.pdf

Terri. (2025). Retrieved 2025, January 22 from https://www.terri.se/en/

Ulrich, K. T., & Eppinger, S. D. (2012). *Product Design and Development*.

Utkurovna, I. M., & ugli, M. K. Y. (2024). Characteristics of Fabrics Used in Designing Modern Backpacks for School Students. *INTERNATIONAL JOURNAL OF FORMAL EDUCATION*.

Van Boeijen, A., Daalhuizen, J., Van Der Schoor, R., & Zijlstra, J. (2014). *Delft design guide: Design strategies and methods*.

Van Gorp, T., & Adams, E. (2012). *Design for emotion*. Elsevier.

Wealth creation. (2024). Retrieved 2025, January 22 from https://www.forestindustries.se/forest-industry/forest-industry-significance/wealth-creation/

Wilson, C. (2010). CHAPTER 4 - Brainstorming. In C. Wilson (Ed.), *User Experience Re-Mastered* (pp. 107-134). Morgan Kaufmann. https://doi.org/https://doi.org/10.1016/B978-0-12-375114-0.00007-4

Yue Wang and Mitchell, M. T. (2011). Integrating comprehensive customer requirements into product design. *CIRP Annals*, *60*(1), 175-178. https://doi.org/https://doi.org/10.1016/j.cirp.2011.03.091

Ørngreen, R., & Levinsen, K. T. (2017). *Workshops as a Research Methodology*. https://www.researchgate.net/publication/317218341_Workshops_as_a_research_methodology

8. Appendix

Appendix

Carry the Forest:

Designing a Bag for Forestry Machine Operators

PAPER WITHIN *Product Design*

AUTHOR: *Sandra Jarauta Abadía*

JÖNKÖPING May 2025

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2. Materials research

Material	Durability	Weight	Weather Resistance	Cost	Environmental Impact
Ballistic Nylon	Very High	Heavy	High (abrasion & water resistant)	High	High (non-recyclable, synthetic)
Cordura Nylon	High	Medium-Light	High	High	High (synthetic, limited recycling)
Polyester (PET)	Medium-High	Medium	Medium-High	Low	Moderate (non-biodegradable)
TPU-Coated Fabric	High (surface layer)	Medium	Very High (fully waterproof)	Medium	Moderate (long lifespan, non-green)
Waxed Canvas	Medium-High	Heavy	Medium (water-resistant, breathable)	Medium-High	Moderate (natural base, needs care)
Ripstop Fabric	High	Light	High	High	High (synthetic, hard to recycle)
Recycled Polyester (rPET)	**Medium-High**	**Medium**	**High**	**Medium**	**Low (made from recycled plastics)**
Recycled Nylon	High	Medium-Light	High	Medium-High	Low (reduced waste, eco-friendly)

Material	Pros	Cons	Best Use Case
Ballistic Nylon	Extremely durable, abrasion-resistant, water-resistant	Heavy, expensive, distort in extreme heat	Heavy-duty, industrial-use backpacks
Cordura Nylon	Durable and wear-resistant, lightweight, water-resistant	Can be costly, stiffer	Outdoor, tactical, work backpacks
Polyester (PET)	UV-resistant, severe frost resistant, inexpensive, lightweight, high strength, significant wear resistance	Less durable than nylon and heavier	Budget-friendly and casual use
TPU-Coated Fabric	Fully waterproof, resistant to chemicals and oil, cost-effective	Less breathable, degradation over time	Industrial and waterproof backpacks, travelling
Waxed Canvas	Stylish, durable, water-resistant, breathable, lightweight	Heavier, requires maintenance	Rugged and eco-friendly backpacks, school bags
Ripstop Fabric	Tear resistant, lightweight, water resistant	Costly, rigid	Hiking gear, military bags
Recycled Polyester (rPET)	**Durable, water resistance, lower waste**	**Microplastic shedding**	**Everyday bags, outdoor bags**
Recycled Nylon	Durable, water resistance, lower waste	Limited availability	Outdoors backpakcs, fashion bags

3. Questionnaire

Survey: Bag for forest machine operators

Hello!

I am a master's student in the product design program at Jönköping University and am currently working on my thesis. My research focuses on designing an efficient way for forest machine operators to transport their personal belongings from home to the machine.

I would be very grateful if you could take about five minutes to answer the following questions. Your answers are confidential and will be used for research purposes only.

Thank you for your valuable contribution to my project – your insights and experiences play a crucial role in this work! :)

1. How old are you? *

- ◯ Under 25 years
- ◯ 25-40 years
- ◯ 41-60 years
- ◯ 60+ years

2. What machine do you drive? (Please state machine type, make, model, year, etc.) *

Your answer

3. Do you bring things from home to the cabin when you drive the forestry machine every day? *

- ◯ And
- ◯ The best

4. How do you usually transport these things from home to your machine every day? *

- ◯ Backpack
- ◯ Shoulder bag
- ◯ Suitcase
- ◯ Plastic bag
- ◯ Toolbox
- ◯ Other: _____

5. Approximately how much does your bag, including its contents, weigh when you go to the forestry machine? *

- ◯ 0-2 kg
- ◯ 2-5 kg
- ◯ 5-10 kg
- ◯ 10+ kg

6. How far do you usually carry your bag from the car to the machine? *

- ○ 0-500 m
- ○ 501 m-2 km
- ○ 2-5 km
- ○ More than 5 km

7. What things do you usually take with you from home to the forestry machine every day? *

- ☐ Along with
- ☐ Drink
- ☐ Extra clothes
- ☐ Tool
- ☐ Electronic devices (mobile phone, tablet, charger, etc.)
- ☐ First aid kit
- ☐ Other: _____

8. What do you like about the bag you use? *

Your answer

9. What do you NOT like about the bag you use? *

Your answer

10. What features do you wish your bag had? *

Your answer

11. Where do you usually place your bag inside the engine cabin? *

- ○ On the seat
- ○ On the floor
- ○ In a storage space
- ○ Suspended on a hook
- ○ Other

12. How often do you need to access your things in your bag during the working day in the cab? *

- ○ Rarely (only during breaks)
- ○ A couple of times a day
- ○ Often (several times per hour)

13. Would you like a specially designed bag to match your forestry machine? *

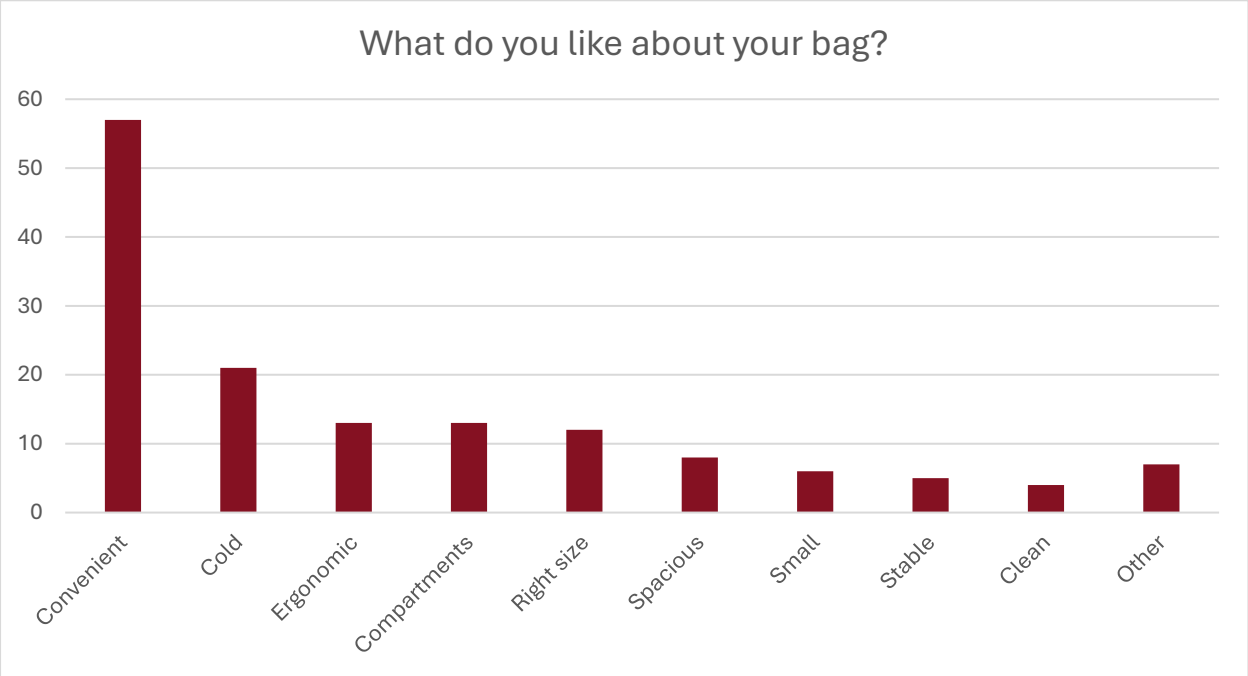
- ◯ And
- ◯ The best
- ◯ No opinion

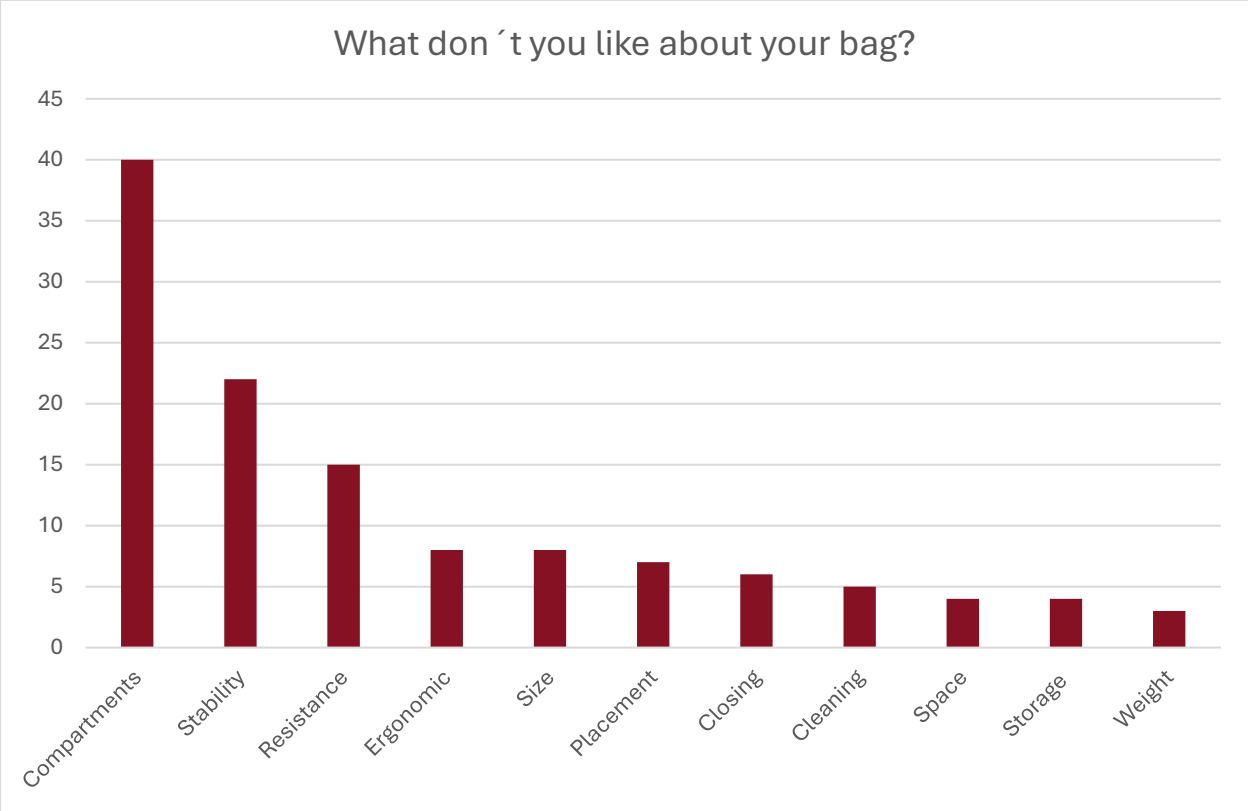
Picture

If you're comfortable with it, I'd love to see how you transport your belongings. It would be very useful for my research.

Could you send a picture of your bag via Facebook, to this account: ...? All pictures will be treated confidentially and used for research purposes only.

Thank you for your answers and your time!


What do you like about your bag?
60
50
40
30
20
10
0
Convenient
Cold
Ergonomic
Compartments
Right size
Spacious
Small
Stable
Clean
Other


What don ´t you like about your bag?
45
40
35
30
25
20
15
10
5
0
Compartments
Stability
Resistance
Ergonomic
Size
Placement
Closing
Cleaning
Space
Storage
Weight

4. Customer statements

	Customer Statement	Translated Interpreted Need	Priority	
Stability	The bag is a little too soft and unstable in itself so it tips over easily.	The bag stays straight by itself	1	must
Ergonomy	The shoulder strap slips off my shoulder if I walk longer distances. It would be good to be able to have it both as a shoulder bag and as a backpack	The bags can be carried on the bag and with the hands	1	must
Closing	It should be easy to open/close. The zipper broke quickly	The bag is easy to open and close	2	should
Compartments	Several storage compartments. Preferably high-built and narrow.	The bag has different compartments for different purposes	1	must
Stability	Tilts easily, must stand flat on the floor, no stable base plate so that the lunch box doesn't tip over so all the fxcking food ends up in the bag	The bag stays straight by itself	1	must
Cleaning	Gets dirty easily, is difficult to clean which can contribute to some odor formation	The bag enables easy cleaning	2	should
Placement	A little awkward to store in the machine and easily falls to the floor and gets in the way like almost all bags do. It's a pain when the bag is open and comes rolling across the floor	The bag has a fixed placement in the cab where it doesn ´t move	1	must
Compartments	No room for a standing thermos	The bag has different compartments where things stay in place	2	should
Ergonomy	A bit heavy to carry when they get longer to walk. Preferably on the back to relieve the arms and be able to hold, for example, a flashlight	The bag provides enables an ergonomic way of carrying it	2	should
Cooling	Should be insulated to keep things cold, it doesn't keep food and drinks cold.	The bag keeps specific items cold	2	should
Compartments	Poorly adapted to bring an iPad and charger, would have been better if it was hard in the shell.	The bag has different compartments for different purposes	2	should
Cleaning	The exterior material is difficult to clean if it gets dirty as it is a fabric that absorbs a lot	The bag enables easy cleaning	2	should

5. Requirements list

No.	Catergory	Requirement	Priority
1	Stability	stays straight	1
2		has a hard base	2
3		has hard walls to stay straight	1
4	Placement	stays fixed in a spot in the cabin	1
5	Ergonomic	can be carried on the back	1
6		straps can be adjusted	1
7		can be carried on the shoulder	2
8		has a strap to carry it with one hand	1
9		enables an ergonomic way of carrying it	1
10	Accesibility	provides easy access to compartments	1
11		allows easy opening	1
12		allows easy closing	1
13	Cleaning	is easy to clean the inside	1
14		is easy to clean the outside	1
15		is water proof inside	1
16		is water proof outside	1
17	Compartments	has different compartments for different purposes	1
18		keeps items stay in place	1
19		has straps to anchor items in place	2
20		has a compartment for electronics	2
21		has a compartment for bottles	1
22		has a compartment for straight bottles	3
23		has a small pockets for small items	1
24		has a cooling compartment	1
25		has a hard compartment for ipad	2
26		has a mesh compartment	3
27		has detachable compartments	3
28		has heavy dutty zipper	2
29		has insulated walls	1
30		has enough space for two lunchboxes	2
31	Resistance	is resistant to tearing	2
32		is resistant to oils	2

1 must-have	mandatory
2 should-have	necessary
3 nice-to-have	unecesary

6. Cooling experiment

Objective: determine the temperature of the drinks after 8 hours shift in a room temperature environment.

Cans volume: 330mml

Can A: INSULATED

Can B: NORMAL

Can C: CONTROL

EXPERIMENT 1

1. 3 cans are placed in the fridge during the night
2. In the morning, they are taken out of the fridge, they are cold, fridge temperature
3. Can A is placed in the insulated bag
4. Can B is placed in the normal bag
5. Can C is opened and the liquid temperature is measured

8 hours pass by

1. Can A is opened and the liquid temperature is measured
2. Can B is opened and the liquid temperature is measured

Compare

1. Can C is opened and the liquid temperature is measured = 8 °C, 9:30h
2. Can A is opened and the liquid temperature is measured = 21 °C, 17:30h
3. Can B is opened and the liquid temperature is measured = 22 °C, 17:30h

EXPERIMENT 2 - Cold water 9:30 – 13:30

Insulated bag

1. Tupper: 15 °C - 18°C
2. Thermos: 14 °C - 17°C
3. Sigma bottle: 9 °C - 17°C

Normal bag

1. Tupper: 15 °C - 18°C
2. Thermos: 14 °C - 18°C

3. Sigma bottle: 9 °C - 17°C

EXPERIMENT 3- Cold water with icepacks 9:40 – 13:40

1. Insulated bag
2. Tupper: 13 °C - 14°C
3. Thermos: 13 °C - 15°C
4. Sigma bottle: 9 °C - 15°C

Milwaukee bag

1. Tupper: 13 °C - 13°C
2. Thermos: 13 °C - 15°C
3. Sigma bottle: 9 °C - 14°C

EXPERIMENT 4- Cold water with icepacks 9:40 – 17:40

Insulated bag

1. Tupper: 13 °C - 16°C
2. Thermos: 13 °C - 18 °C
3. Sigma bottle: 9 °C - 17°C

Milwaukee bag

1. Tupper: 13 °C - 16°C
2. Thermos: 13 °C - 15°C

Sigma bottle: 9 °C - 17°C

7. User test

Mock up 1	User 1	User 2	User 3	User 4
Lunch box compartment left	It has to be exactly that meassurement to be possible to grab	Good	Difficult to grab	Good
Lunchbox compartment right	The lunchbox needs something so it can be grabbed	Good	Had to remove the hand rest for easier accesiblity	Good
Insulated compartment, from the top (1,5 l water bottle)	Easy to reach	Very accessible	**It is better if the compartment lid opens completely**	To grab a heavy bottle from the top is not very stable, it can slip
Main compartment (socks)	Good	Good	Good	Good
Main compartment (toilet paper)	For wider people it might be more difficult to reach	Good	**Good but things are stacked, maybe small walls with velcro**	A bit too deep
Maintain main compartment open	Good	**If I leave it open it migh get dirty**	Good	Good
Tablet compartment	Good	**Not very easy to reach, maybe in the front might be easier. With a rope so it goes up**	Good but difficult to have a good grip with the tablet	**Tablet compartment doesn't need to go all the way up**
General feedback	**Good**	**Good**	**Good**	**Good**

Mock up 2	User 1	User 2	User 3	User 4
Lunch box compartment top	Depends on the size	Better	Better like this	Coulmn of the wall is in the way
Lunchbox compartment bottom	Depends on the size, no big difference	**Better. The top compartment is more accessible, could be for fika, more than once a day**	Same	Good
Insulated compartmen, from the side (1,5 l water bottle)	Perfect	**Compartment doesn't need to be that big for one water bottle. Have another compartment for less used things**	**Easier if it would open with an angle**	Good
Small top compartment (snus)	Easier than previous one	Good	Easier	Better
Main compartment (toilet paper)	Easier, more accesible	Better, less deep	Better	Good
Maintain main compartment open	Same	Same	Same	If there is light that lits up the bag
Tablet compartment	Same	Same	Same	Good
General feedback	**It is more accesible and easier. Better reachability**	**Better, more reachability**	**Better**	**Better**

CONCLUSIONS		
Mock up 2 is better because:	The water bottle is easier to access from the side, there is more space	It can be open with an angle, To grab a heavy bottle from the top is not very stable, it can slip
	The lunchbox compartments are close. One can be for easier access like snacks or fika	Give some extra space for the grip. The are not together because you would need two hands
	Top compartment is devided so it has different functions	It can have a velcro wall that can be removed
	Top big compartment is less deep	
	Top small compartment can be for smaller items	With some mesh